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Product Life Cycle Information as an Enabler for Circular Practices

A Case Study

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Abstract

This report is a master's degree project in Integrated Product Development at KTH Royal Institute of Technology, conducted in collaboration with Knightec Group and a case company.

The case company, an international hose assembling company, has experienced challenges with identifying and integrating circular practices into their operations. Through a case study approach, the key opportunities and critical enablers for a successful transition towards circularity were identified. The opportunity selected to investigate further was that of how life cycle information of the case company's products can enable circular practices at the case company.

Following a literature study, desktop research, qualitative interviews and site visits, a Life Cycle Information Gathering Process was developed. The process answers the questions of *what* life cycle information should be gathered, *how* it should be gathered, *who* should gather it and *when* it should be gathered in five selected life cycle stages of a product. The process is a support to guide the case company to integrate life cycle information gathering to a greater extent into current operations. Opportunities and risks of implementing the process have been analyzed. Moreover, recommendations on further actions that the case company should implement to transition towards circularity are detailed.

Collecting life cycle information was identified to enable circular practices such as enhanced waste management, lifetime estimates of products and increased servitization opportunities for the case company. Furthermore, the report contributes to the limited research on life cycle information as an enabler for circularity and its collection process.

Keywords

Information Gathering, Circular Economy, Information Flow, Collection Process, Ways of Working

Sammanfattning

Denna rapport är ett examensarbete på masternivå inom Integrerad Produktutveckling vid KTH Kungliga Tekniska Högskolan, genomfört i samarbete med Knightec Group och ett fallföretag.

Fallföretaget, ett internationellt företag som sammansätter slangar och ledningskomponenter, har mött utmaningar med att identifiera och integrera cirkulära arbetssätt i sin verksamhet. Genom en fallstudie identifierades ett antal centrala möjligheter och möjliggörare för en lyckad omställning mot cirkularitet. Den möjlighet som valdes för vidare undersökning var hur livscykelinformation av fallföretagets produkter kan möjliggöra cirkulära arbetssätt i organisationen.

Med utgångspunkt i en litteraturstudie, skrivbordsforskning, kvalitativa intervjuer och platsbesök utvecklades en Process För Insamling Av Livscykelinformation. Processen besvarar frågorna om *vilken* livscykelinformation som bör samlas in, *hur* den ska samlas in, *vem* som ska samla in den och *när* den ska samlas in i fem utvalda faser av produktens livscykel. Syftet med processen är att agera som ett stöd för fallföretaget i deras arbete med att strukturera och genomföra insamling av livscykelinformation. Möjligheter och risker med att implementera processen har analyserats. Därtill ges konkreta rekommendationer på åtgärder som bör vidtas för att möjliggöra en cirkulär omställning.

Att samla in livscykelinformationen identifierades som en möjliggörare för ytterligare cirkulära arbetssätt såsom förbättrad avfallshantering, uppskattning av livslängd på produkter och ökade möjligheter till tjänstefiering på fallföretaget. Rapporten bidrar även till den idag begränsade forskningen kring livscykelinformation som en möjliggörare för cirkularitet och dess insamlingsprocess.

Nyckelord

Informationsinsamling, Cirkulär Ekonomi, Informationsflöde, Insamlingsprocess, Arbetssätt

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Stockholm, June 2025
Emma Jakobsson and Emeline Veillas

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List of acronyms and abbreviations

CE	Circular Economy
BM	Business Model
EU	European Union
LCI	Life Cycle Information
LC	Life Cycle
HQ	Headquarter
EGD	European Green Deal
CEAP	Circular Economy Action Plan
ESPR	Ecodesign for Sustainable Products Regulation
DPP	Digital Product Passport
OEM	Original Equipment Manufacturer
PIM	Product Information Management
ERP	Enterprise Resource Planning
IMDS	International Material Data System
BOM	Bill of Materials
LCA	Life Cycle Assessment
LCI	Life Cycle Inventory
SRM	Supplier Relationship Management

1 INTRODUCTION

This report presents a master's degree project in Integrated Product Development at KTH Royal Institute of Technology in collaboration with Knightec Group and a case company. This chapter provides the background to the problem that this thesis addresses, the project partners, the project brief and delimitations made.

1.1 Background and Problem Description

Human activity constitutes the most influential contributor to changes of the Earth's climate (Rockström et al., 2009). Europe is continuing to consume more resources, which is causing environmental degradation and problems in several areas, including biodiversity loss, climate change and health risks (European Environment Agency, 2024). Concerns of how human activity can trigger permanent environmental changes, in other words cross the planetary boundaries, have been raised (Rockström et al., 2009).

To help reduce environmental pressures, the concept of Circular Economy (CE) is viewed as a suitable sustainable solution (Ellen MacArthur Foundation, n.d.; European Commission (b), n.d.). The idea is to rethink the current linear economic process by adopting an alternative circular flow model, decoupling economic growth from resource consumption (Ellen MacArthur Foundation, n.d.; Korhonen et al., 2018). Essentially, the concept revolves around maximizing and retaining the value of products, parts and materials through their recovery and reuse, either extending their lifetime or feeding new product loops (Ellen MacArthur Foundation, n.d.). The system also focuses on reducing consumption and using resources more consciously to minimize waste and pollution (Uwuigbe et al., 2025).

To accelerate the transition towards CE, legislation and regulations are key drivers (de Mattos & de Albuquerque, 2018). With an established proposal, named the Circular Economy Action Plan (CEAP), the European Union (EU) is among those who envision such a transition (European Commission (b), n.d.). As a constituent part of the European Green Deal (EGD), this strategy presents a set of initiatives and targets to achieve sustainable growth and decreased environmental pressures (European Commission (a), 2021). To reach such objectives, the Ecodesign for Sustainable Products Regulation (ESPR) suggests a collection of measures in the form of a regulatory framework, specifying performance and information rules for almost all physical goods categories (European Commission (d), n.d.).

A specific measure taken in the ESPR is the implementation of Digital Product Passports (DPP). This tool acts as a digital identity card for products, collecting and tracking product life cycle information, to allow for more sustainable decisions across all product phases (European Commission (d), n.d.; Lopes & Barata, 2024; Walden et al., 2021).

For a company transitioning to a circular model, there are multiple challenges and components to consider. Among others, the ability to track products, components and materials is a key element when constructing a CE (de Mattos & de Albuquerque, 2018). As promoted by the EU through the upcoming DPP regulation, product information flow is significant for circular practices and more sustainable production and consumption (Adisorn et al., 2021). However, collecting life cycle information across all phases of a product's life is challenging (Lopes & Barata, 2024), particularly during the use phase (Adisorn et al., 2021). For example, managing the interests and interactions among involved stakeholders, along with balancing data access and privacy requirements, is complex (Lopes & Barata, 2024).

Companies are facing an increased pressure to become more aware of their environmental footprint and take responsibility (Hall & Vredenburg, 2003). Among these companies are incumbent firms, those already established in the industry. Due to their size, incumbents have a better ability to achieve large scale impact in terms of sustainability (Geels, 2011). While incumbents are putting a priority on sustainability issues, there are challenges connected to transitioning from linear to Circular Business Models (CBMs). Successful transformation requires a shift in how incumbents operate, affecting how they are creating, delivering and capturing value. Additionally, there is a lack of systemic processes for such transformations (Frishammar & Parida, 2019).

Driven by upcoming regulations and legislation, the transition towards CE for companies, especially incumbents, is a highly relevant field of research. There is a need to examine how and if product life cycle data collection can enable the start of such a transition. This creates an opportunity for exploring the importance and impact of product life cycle data on an incumbent's journey towards becoming circular and how it enables sustainable practices.

1.2 The Project Partners

The employing company for this thesis is Knightec Group, a consulting company with clients across multiple industries, ranging from the medical industry to automotive and manufacturing. With a strong focus on innovation and transformation of businesses, Knightec Group offers consulting services within circularity to, among other goals, reduce environmental impact of business operations.

The client of the project will be anonymous in this report and referred to as *the case company*. The case company offers services and solutions for hoses, fittings and related products. Since founded, the case company has established operations globally with a large number of stores, mobile servicing vans and resellers. The case company offers a massive range of products, offering over 20.000 items in their product catalogue. Considering the scale of the company and product assortment, the case company has a wide variety of customers, ranging all the way from private customers to large corporations, where they often act as an Original Equipment Manufacturer (OEM). The large turnover of products also states the importance for the case company to take responsibility for the environmental impact they imply.

The case company offers so-called hose assemblies for both high-pressure and industrial application areas. The assemblies can consist of hoses, fittings, adapters, ball valves, mounting technology and swivels, depending on application. See an example of a hose assembly in Figure 1.1. The case company's specialty is assembling these components in a durable way. They also offer products within the fields of pneumatic technology, pipes, measurement equipment and crimping machines.



Figure 1.1: An example of a hydraulic hose assembly.

The case company has begun their journey towards sustainability and has implemented some strategies to move towards circularity. Two sustainability reports for the years 2023 and 2024 have been published, as well as an established sustainability policy. However, the case company still faces several challenges in reducing their environmental impact and improving sustainability. A complex product assembly, material mixes and lack of knowledge regarding the product's life cycle complicates product end-of-life opportunities. This has been recognized by the case company, and these complexities lay the foundation for this thesis.

1.3 Brief

The brief of this thesis was to investigate, identify and map out possible solutions to enable circular practices at the case company. The goal is to provide guidance to assist the case company in their transition towards circularity.

Given the broad brief, multiple possible project scopes were explored in a first pre-study phase. Various focus areas were proposed and discussed with the case company supervisors to narrow down the scope of the project. Figure 1.2 shows the narrowed down scopes introduced to the case company.

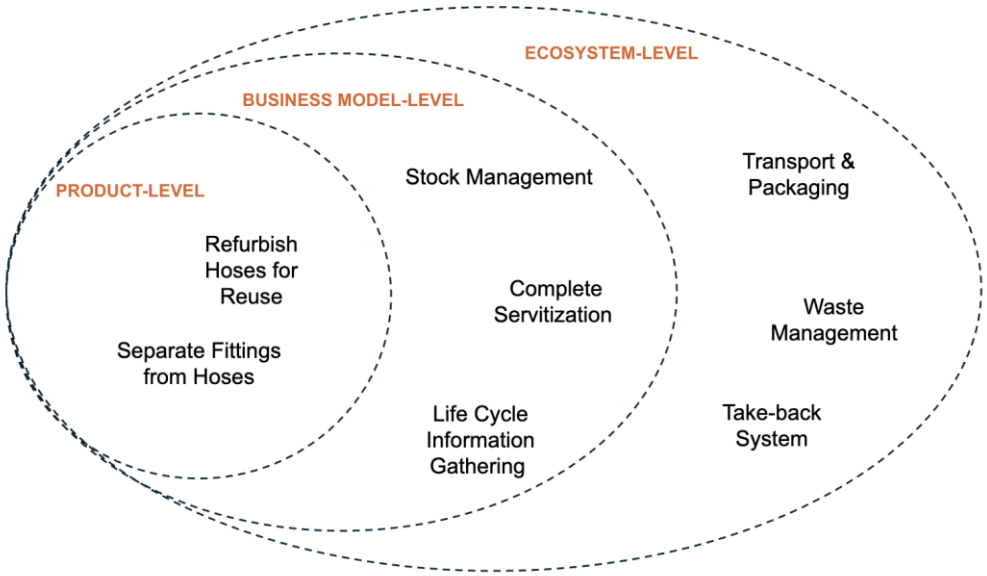


Figure 1.2: Possible scopes for the project divided into perspective levels.

Through careful evaluation and discussions with industrial and academic supervisors, attaining information on product life cycles was chosen as the topic for this thesis project. Product life cycle information was identified as a crucial enabler for multiple circular practices for the case company. For example, other project scopes, such as enhancing waste management, can be facilitated if more product life cycle information is available. Furthermore, it was also central to investigate how companies transition towards circularity in practice and how this influences companies' ways of working.

1.4 Purpose and Problem Definition

On a broader perspective, the purpose of this master thesis is to explore actions necessary for the case company to transition towards circularity. More specifically, the purpose is to investigate how information gathering on products' life cycles can enable circular practices for an international hose assembling company. The focus lies on information as an enabler for sustainability and a crucial aspect to permit circularity. Furthermore, the goal is to explore what information collecting solutions are implementable for an international hose assembling company. The study includes examining opportunities and risks that come with life cycle information gathering.

To guide our study, three specific research questions were formulated. The first research question was defined to understand the connection between product life cycle information and circular practices. The second question aimed to support the exploration of a solution to tackle life cycle information gathering. Finally, the last question assisted the analysis of the developed solution. They read as follows:

1. How can information gathering on products' life cycles enable circular practices for an international hose assembling company?

2. What information gathering solutions are valuable to enable circular practices for an international hose assembling company?

3. What opportunities and risks would come with such a solution?

1.5 Delimitations

To achieve a valuable and actionable project scope, delimitations have been made throughout the thesis project. The foremost prerequisite is the project time frame of 20 weeks, restricting the research scope. Furthermore, the research area focuses solely on the role of information and data collection practices, meaning that the effects after implementation of such a solution is not explored in depth. However, initial analysis of possible effects, recommendations and value of implementation is examined. Additionally, no exploration of possible technologies and systems for gathering and storing life cycle information is done in this thesis.

As sustainability incorporates multiple perspectives, and involving all these aspects would require extensive research, the authors decided to narrow down this aspect of the scope. It was chosen to focus on environmental aspects of sustainability but also incorporates limited views of the economic and social factors.

The case company offers a wide variety of products. Material composition, product architecture and target customers vary immensely between the product categories. Therefore, the thesis project is limited to focus on industrial and high-pressure hoses as well as their associated fittings.

The customer segments for industrial and high-pressure hoses vary from large corporations to private customers. To create an actionable investigation, the thesis project focuses on the segment of private customers and small business to business sales.

The case company operates globally. However, the thesis is limited to the region of Sweden with respect to the time prerequisite of the project. Therefore, research and data gathered was performed with regards to this limitation.

2 PROCESS AND METHODOLOGY

This chapter describes the process and methods used during the thesis project. The methods are presented under each phase of the process.

2.1 Process Description – Triple Diamond

The process for this thesis is based on the Double Diamond framework. Launched in 2004 by the Design Council, the framework describes key principles and design methods for an innovative design process. The framework allows designers to alternate between divergent and convergent thinking, allowing for both exploration and focused actions (*Framework for Innovation - Design Council*, n.d.). Overtime, the framework has evolved and changed to better fit a bigger mix of problems. The Triple Diamond Design Process expands the original model from two to three diamonds. This expanded model is designed to provide a holistic approach to solving the problem at hand through 6 phases: Dialogue, Diagnose, Discover, Define, Develop and Deliver (*The Triple Diamond Design Process*, 2024). Due to the broadly stated brief of the thesis project and the demand for a holistic perspective, the Triple Diamond Process has been selected. The dialogue and diagnose phase allow the authors to explore what opportunities there are for solving the problem stated in the project brief, before narrowing down a specific scope and exploring it further in the remaining phases. Furthermore, the authors decided to divide the triple diamond further into two parts to facilitate project planning. The first part aimed to explore sustainability issues in general at the case company, while the second entailed targeted exploration and development of the selected scope. These phases are illustrated in Figure 2.1 and explained in sections 2.2 to 2.7.

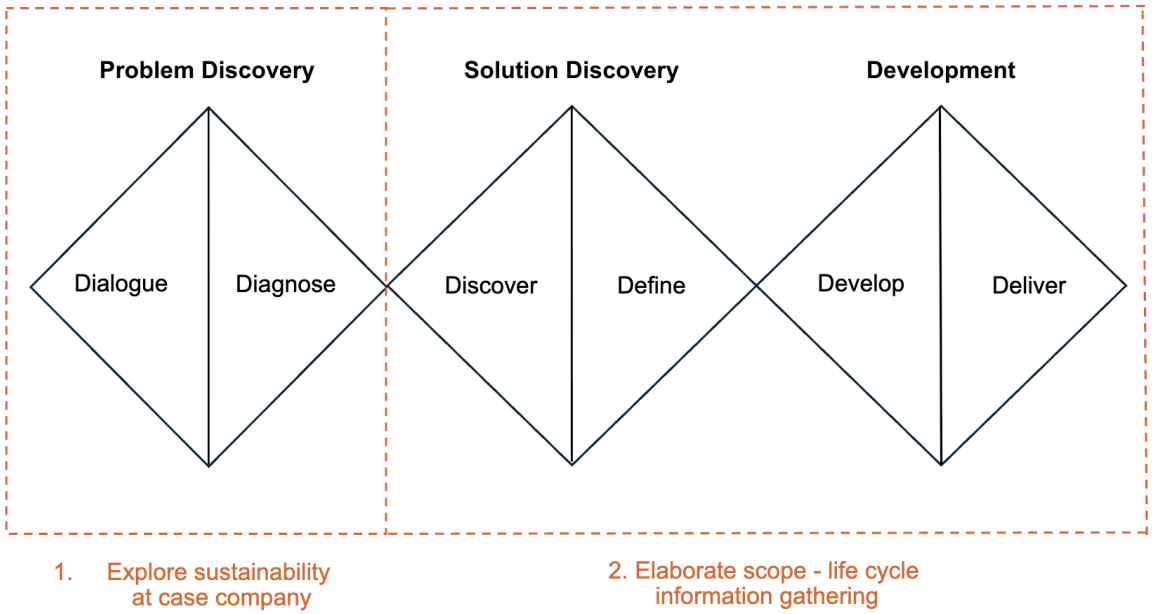


Figure 2.1: Illustration of the Triple Diamond Process and our process.

2.2 Dialogue

In the Dialogue phase, the main purpose is to explore and gain extensive knowledge of the problem space. By exploring the context of the problem, potential pains, challenges and opportunities are mapped out (*The Triple Diamond Design Process*, 2024). In this thesis, the problem description given

by the case company - improving sustainability through product development - was broad. Therefore, extensive exploration of the company's current practices, product offering, organizational structure and similar, was essential before a narrower problem description could be formulated. This phase meant to deeply analyze current sustainability practices at the case company and map out future opportunities and current threats related to these. The methods used in this phase are listed below.

2.2.1 Industry Analysis and Research of Related Industries

To identify potential problems and opportunities for the case company, knowledge of the industry had to be attained first. This was done through desktop research on the hose industry as well as related industries and products, such as tires and cords. This provided insights into how other industries handle the topics of sustainability, including topics such as waste management, reuse and recycling, of products with similar material mixes to the case company products. The results of this research were documented in the digital whiteboard Miro.

2.2.2 Product Mapping

To gain knowledge of application areas, characteristics and material mixes of the case company's products, a product map was created. Originating from the main product categories of the case company, related products to these were mapped out and categorized accordingly. The product map was an essential base for understanding the case company's product offering and provided guidance during meetings when discussing which product category to focus the thesis on. Also, it was beneficial during ideation when circular design possibilities of different products were investigated. The product mapping was done in Miro.

2.2.3 Study Visit to Case Company Store and Test Lab

A study visit to a case company store provided insights into the everyday operations. An on-site informal interview with an experienced store employee allowed for problem discovery and deeper understanding. Notes were taken during the visit and later utilized as a base for ideation. At the case company headquarters the authors also visited their testing facility, providing insights into product characteristics and demands on function.

2.2.4 Discussion Meetings

To facilitate the initial understanding of the case company, industry and products and the sustainability challenges at the case company, a series of informal semi-structured meetings were performed with the industrial supervisors at the case company. Due to the lack of written documentation on these subjects, the informal meetings were essential to obtain such information. The industrial supervisors, a product manager and the sustainability manager at the case company, possessed knowledge on both the product level and overall sustainability issues, assisting the authors with their understanding of the case company.

2.2.5 Stakeholder Journey Map

Stakeholder journey mapping was essential for identifying which stakeholders are directly or indirectly affected by the case company operations. The map was created by analyzing which stakeholders the case company are in contact with from the beginning of the products journey until the end. It was important for the thesis project to identify which stakeholders would be affected by actions that enable sustainability at the case company. Both an internal and external stakeholder journey map was created in Miro.

2.2.6 Free Brainstorming

The authors engaged in several initial brainstorming sessions, where ideation of possible problems to look further into were brought forward. During these sessions, the authors aimed to think broadly and not restrict themselves in bringing forward ideas for different scopes to focus on. The brainstorming exercises were first documented physically on post-it notes and later digitized in Miro.

2.2.7 Workshop - Circularity Deck

The Circularity Deck workshop is a tool designed to help understand and identify circular and sustainable business practices that could be implemented in a company (Konietzko et al., 2020). The illustration used for the workshop is presented in Appendix A. The authors utilized this tool to ideate on possible solutions enabling sustainability. This workshop was done twice, once by the authors alone, and once together with industrial supervisors at the case company. The reason for conducting the workshop twice was to look for similarities and differences in the resulting ideas. This aided the authors to select ideas which were deemed the most feasible and valuable for the case company. The resulting ideas were documented physically on post-it notes and later digitized in Miro.

The result of the dialogue phase was a wide range of possible scope ideas to focus the thesis on and a solid base for the remainder of the project. These were documented in Miro and laid the foundation for the upcoming diagnose phase.

2.3 Diagnose

With gained knowledge from the dialogue phase, a specific challenge is chosen to be investigated further in the Diagnose phase. At this stage, the scope and a narrower problem description of the thesis was formulated and provided a basis for the upcoming phases. The methods used in this phase are listed below.

2.3.1 Opportunity Mapping for Scope Evaluation and Selection

In the dialogue phase, eight scope opportunities were identified. The authors proceeded to identify opportunities and threats for each of them. Important factors considered in the opportunity mapping were ease of implementation, financial investments needed, sustainability impact and feasibility. This was done through verbal discussions with the industrial supervisors and documented in Miro. This facilitated narrowing down the number of possible scopes to the one with the most potential for improving sustainability long-term.

2.3.2 Scope Meeting

After the opportunity mapping, five different scopes were selected to move forward with. The decision was based on the requirements that the selected scope should be easily implementable into current operations at the case company, as well as making the most impact on a successful transition towards circularity. The final selection was made during discussion meetings with both academic and industrial supervisors. Their insights aided the authors to decide which scope to select for the thesis.

2.4 Discover

The Discover phase encourages diverging and taking a broad look at the specific problem at hand. Information is gathered by employing an open mindset and deep understanding of users' perspectives (Kochanowska et al., 2022). In this step, the challenge is explored once more through empirical and literature research, with the goal to formulate potential solutions. The methods used in this phase are listed below.

2.4.1 Literature study

A literature study was conducted to establish a theoretical base for the chosen scope. The databases Scopus and Google Scholar were utilized to retrieve relevant literature used in this thesis. The literature retrieved focused on circularity, CE, circular strategies, circular transitions for incumbents, sustainable development, information gathering of product life cycles, EU regulations for circularity and processes for circular transitions.

The authors identified that research on the topic of information was difficult, as it is a word commonly used in other contexts than that of life cycle information. The keywords shown to be the most successful for attaining useful articles on this topic were the following: "Information", "Information collection", "Information gathering" and "Product Life Cycle".

The papers were chosen by evaluating the publishing journal, number of citations and relevancy to the thesis. Once a few relevant articles had been identified, the Snowballing method was used to identify additional relevant literature. Snowballing is a widely used method in literature research to localize articles in the same or similar field by exploring reference lists and citations of the initial paper (Wohlin, 2014). The articles used were saved in the reference management tool Zotero, facilitating collecting, annotation, organization and citation of chosen articles. This process was done iteratively until enough articles were retrieved. During the Develop phase, an additional in-depth literature study was conducted to retrieve specific literature relating to the result developed in the thesis.

2.4.2 Interviews

Interviews were chosen as the main data gathering method for this thesis as it was deemed to be the most efficient way to retrieve information on current practices, opinions and issues at the case company. The interviews were semi-structured which allowed for in-depth information to be attained and flexibility as the interview progressed (Ruslin, 2022). The semi-structure of the interviews meant that the authors prepared topics and questions deemed fitting for the role of the interviewee, while allowing for deviation and deeper discussions on topics brought forward by the interviewee. Therefore, a unique interview guide was created for each interview. A generalized interview guide can be found in Appendix C. This was beneficial as it allowed the authors to uncover new ideas and aspects based on the interviewee's expertise. A total of 14 interviews (not including informal interviews presented in previous phases) were conducted continuously throughout the thesis, most of them were held in the Discover phase. An overview of the interviewees held is shown in Table 2.1, presenting the role of the interviewee, whether they are employed at the case company or an external stakeholder, and the interview duration.

Table 2.1: Overview of interviewees in chronological order.

Role of interviewee(s)	Internal/External	Duration [min]
Logistics coordinator & Manager of quality and environment	Internal	45
Site Manager OEM	Internal	75
District Manager	Internal	58
Supply Chain Manager	Internal	76
Group Marketing Manager	Internal	66
Waste Management Company Manager	External	60
Market and Customer Manager	Internal	47
Supplier Quality Developer & Group Sustainability Manager	Internal	60
Technical Customer Support	Internal	48
Sales - High Pressure	Internal	58
Service Manager – Mobile Hose Service	Internal	84
Concept Manager - Mobile Hose Service	Internal	44
Systems Owner	Internal	60
Sales - High Pressure (follow up interview)	Internal	40

All interviews were held digitally via Teams, recorded and transcribed after consent from interviewee. All interviews were held in Swedish, citations used in the report and their translation are presented in Appendix C to ensure transparency.

2.5 Define

In the Define phase, the data collected in the previous phase is analyzed. According to Kochanowska (2022), it is important to look for patterns in findings and understand stakeholder needs. From these insights, a solution concept is formulated (Kochanowska et al., 2022). In this phase, multiple solutions were formulated and evaluated. In the end, one solution was selected to develop further in the third diamond. The methods used in this phase are listed below.

2.5.1 Clustering - Analysis and Results from Interviews

To effectively document and analyze the content of the interviews, the authors created a codebook consisting of 15 codes to categorize the content. The codebook was created according to a method developed by DeCuir-Gunby et al (2011). The purpose of analyzing the interview data was to answer the thesis projects research questions, see section 1.4. Therefore, the authors choose to develop structural codes, meaning that they originated from the research questions (DeCuir-Gunby et al., 2011). The codebook consists of a code name and a brief definition of what is referred to, presented in table 2.2. Furthermore, the codes were categorized into code groups, subthemes and large themes, as illustrated in figure 4.1. This structure was beneficial for grouping the interview data and presenting it in a logical way for the reader.

Table 2.2: Codes used for interview analysis and explanation.

Code	Brief Definition
General life cycle analysis	Interviewee refers to life cycle data in general.
Data collection - Type of data	Interviewee refers to life cycle data collection of hoses or other products. What kind of data?
Life cycle data gathering technologies	Interviewee refers to life cycle data collection of hoses or other products. What kind of technologies are used for gathering data?
Life cycle data gathering solutions	Interviewee refers to life cycle data collection of hoses or other products. What kind of solutions are used today? Are ideas for new solutions mentioned?
Opportunities with life cycle data	Interviewee point out benefits or opportunities related to collecting life cycle data. E.g. benefits for waste management.
Challenges and risks of gathering life cycle data	Interviewee refers to challenges and risks of gathering life cycle data. E.g. difficulties in establishing new routines, company secrets etc.
Attitudes towards and incentives to gather life cycle data	Interviewee mentions thoughts on the incentive for gathering life cycle data, can be both positive and negative.
Sustainability or circularity goals and definitions	Interviewee mentions sustainability or circularity goals and definitions for the case company.
Sustainability or circularity practices	Interviewee refers to sustainability or circularity actions taken by the case company. E.g. practices and initiatives.
Challenges of implementing sustainability	Interviewee points out challenges for implementing sustainability initiatives. E.g. organizational resistance, economic limitations, etc.
Sustainability as a competitive edge / marketing advantage	Interviewee points out if and how increased sustainability initiatives affect their competitive placement. E.g. customers appreciate the sustainability report, or similar.
Attitudes towards sustainability and circularity	Interviewee refers to attitudes towards sustainability and circularity. E.g. Have attitudes shifted through the years?
Best practice, ways of working, processes for life cycle information gathering	Interviewee refers to ways of working, best practices and processes for life cycle information gathering at the case company. E.g. How does the mobile hose service work information gathering?
Challenges with current ways of working connected to life cycle information gathering	Interviewee refers to challenges with the way they work, which might also be connected to how this could complicate life cycle information gathering. E.g. employees experience lack of time for these tasks.
Needs related to ways of working with life cycle information gathering	Interviewee refers to needs connected to working with life cycle information gathering. E.g. the working process needs to be time efficient.

By utilizing the codes stated in Table 2.2, the authors were able to effectively organize interview data and gain understanding of the results. How the codes were utilized will be explained in further detail in Chapter 4.

2.5.2 Formulation of Problem Definition used for Developing the Concept

After all data had been gathered and analyzed, the authors had a solid foundation for stating a problem definition and evaluating the most suitable solution to this. The decision to develop a life cycle information gathering process was taken. This laid the foundation for the last part of the triple diamond.

2.6 Develop

In the Develop phase, creative ideation on the selected solution to the problem is performed. Knowledge and understanding of the needs established in previous phases can be tested. As this is a divergent phase, there should be an openness to several potential solutions to the problem at hand. This allows for a second iteration of the selected concept, identification of potential flaws, improvements or even further opportunities. The methods used in this phase are listed below.

2.6.1 Product Journey Map

The authors analyzed the different product journeys existing at the case company. Depending on the scenario, the product travels through different life cycle stages and diverse stakeholders are involved in the journey from supplier to end-customer. Therefore, it was important to identify and select which life cycle stages to focus on. The product journey map was documented in Miro and is explained further in section 5.2.1. Finally, a decision regarding which product journey scenario to focus on was made.

2.6.2 Competitor analysis

Competitor analysis of the case company's direct competitors was done to gain insight into actions taken to enable sustainability within the same or similar industries. The analysis was done through desktop research of the competitors. It also helped identify the case company's competitive position within sustainable practices. Specific attention was paid to competitors who have implemented life cycle information gathering processes in their operations. The results were documented in Miro.

2.6.3 Study visit – Mobile hose service and case company store

To effectively develop the process, a second study visit to a case company store was conducted. Informal interviews with several employees, investigating their needs but also their attitudes and opinions on a life cycle information gathering process, provided additional data to take into consideration in the Develop phase. A ride along with two service technicians working with the mobile service unit, also provided insights into the needs of employees and what requirements the process must fulfill to fit seamlessly into the case company's operations. Further details of these visits can be found in section 5.2.2.

2.6.4 Ideation

Following the Define phase, the authors were able to ideate on what the process of information gathering should contain and how it will be performed. After this initial ideation, a collaborative workshop was performed with the case company. The workshop was designed together with industrial

supervisors at Knightec Group. The purpose of the workshop was to brainstorm ideas around information gathering and generate possible solutions with input from experts at the case company. To gain the most fruitful result, people with different roles at the case company were included. The workshop utilized the concept of “How might we?”-questions to facilitate idea generation. The full description of the workshop can be found in Appendix B. Further details on participants and the workshop agenda can be found in section 5.4.1.

Following the workshop, the authors were able to synthesize the results and brainstorm on their own to develop criteria for the solution. To do this, the authors categorized the findings into the four questions:

- *What?* - What life cycle information should be gathered?
- *How?* - How should the information gathering be done?
- *Who?* - Who is responsible for gathering life cycle information, and who is affected by the process?
- *When?* - In which life cycle stages should life cycle information be collected?

2.6.5 Analysis of Stakeholder Needs

To further investigate what factors affect life cycle information gathering stakeholder needs in the selected scenario were investigated. Similarly to the product journey map, the type of customer will influence the design of the concept. Therefore, the authors utilized data from interviews and the study visit to categorize different types of customers and their needs. Furthermore, the authors discovered that employee experience and type of service job affects the process of gathering life cycle information. Accordingly, classifications were made to understand the variation of needs of these actors. The categorizations can be found in section 5.3.2.

2.7 Deliver

In the final phase, Deliver, the end-solution is processed and refined through testing and iterations. The final delivery is a specific concept with an implementation plan (Kochanowska et al., 2022). The methods used in this phase are listed below.

2.7.1 Value mapping and Risk Analysis

After synthesizing data retrieved so far, the authors were able to identify and map short- and long-term values of information gathering. The value mapping focused on what environmental, social and economic value the case company could expect if the life cycle information gathering process would be implemented. The values were categorized over a timeline ranging from short- to long-term value. Short-term value was defined as value that can be expected to be seen almost immediately after launching the solution, while long-term value would require more resources and time.

2.7.2 Final process and Recommendations

After all the previous phases, a result was developed. Life cycle phases taken into consideration were selected from the chosen product journey scenario. Additionally, answers to the questions of *What*, *How*, *Who* and *When* were formulated. Overall, an information gathering process was developed alongside recommendations to enable circular practices at the case company. Expected results as well as implementation challenges were analyzed in this phase.

3 THEORETICAL FRAMEWORK

This chapter provides a theoretical background connecting sustainability and life cycle information. It describes the following topics in detail: sustainable development, circular economy, circular strategies, laws and regulations, incumbents transitioning and the role of information.

3.1 Sustainability As a Concept

The concepts of sustainability and sustainable development are fundamental in discussions of the environmental, economic and social developments in the world. As environmental pressures on the Earth are increasing, it's now more important than ever for businesses to change the way they operate to comply with sustainable development. Increased awareness of these issues are essential in order to relieve the strain being put on our planet (Bocken et al., 2014).

The terms sustainability and sustainable development carry interrelated meanings, however, in the context of corporate sustainability work they are important to distinguish. Sustainability refers to the phenomenon of balancing economic, social and environmental aspects, creating a state of equilibrium in nature (Mebratu, 1998). Human activity needs to be managed in order to avoid transgressing planetary boundaries, creating irreversible environmental changes to the Earth (Rockström et al., 2009). Sustainability as a term has been used since the early 18th century, producing a number of different definitions in literature since (Geissdoerfer et al., 2017). Over time, a dominant conceptualization of sustainability has boiled down to the idea of three independent pillars: economic, social and environmental sustainability (Purvis et al., 2019). Figure 3.1 illustrates a common illustration of the three-pillar model as a Venn diagram, symbolizing the need to consider all the three perspectives (Purvis et al., 2019).



Figure 3.1: Three intersecting circles, typical representation of sustainability.

By considering these three different pillars, a holistic approach to achieve sustainability is promoted. The social dimension includes the ability to provide both basic human needs, such as access to food, as well as a sense of security and freedom. The environmental dimension promotes the protection of nature and ecological processes, while the economic dimension entails the limitations being put on economic growth in respect of the other two dimensions (Brown et al., 1987). Despite the widespread use of the model, the three-pillar framework fails to articulate how the dimensions interact, especially

when trade-offs occur, causing challenges for organizations when applying sustainability in practice (Purvis et al., 2019).

In contrast to sustainability as a goal, sustainable development describes the dynamic processes of moving towards that goal - achieving a desired state of balance. The most widely cited definition of sustainable development originates from the Brundtland Report: *"development that meets the environmental, social and economic needs of the present without compromising the ability of future generations to meet their own needs*". This definition highlights the balance between social, economic and environmental aspects, and how changes in one pillar can create consequences in another (Brown et al., 1987). Since then, sustainable development has been further systematized and integrated into international agreements. One example is the United Nations Sustainable Development Goals, outlining 17 goals for achieving sustainability by 2030 (United Nations, n.d.).

The purpose of sustainable development is to create an action-plan and establish a common language for addressing global challenges. All members of society should consider sustainable development, among those are companies. By encouraging responsible resource usage, stakeholder collaboration and setting long-term goals, companies can enhance their resilience towards shifts in the ecosystem and work proactively (Ortiz-de-Mandojana & Bansal, 2016). However, sustainable development has also been criticized for its conceptual vagueness. The definition from the Brundtland Report allows for wide interpretation. The absence of a strict definition and practical guidelines creates an uncertainty in how to change in practice in order to be truly sustainable (Purvis et al., 2019).

In response to the uncertainty of implementation of sustainable development, the CE framework emerged as a more tangible approach. The next section introduces the CE concept and its key principles.

3.2 Circular Economy as a Solution for Sustainable Development

To address sustainability issues, comprising environmental pressures, societal problems and economic challenges, there is an urgent need to transition towards a more sustainable system. A concept that has recently gained more and more popularity in the scientific community and professional sector is the Circular Economy. This system approach is seen as a promising solution to achieve sustainable development by moving from a linear to a CBM (Korhonen et al., 2018).

The connection between sustainability and circularity is not crystal clear. Geissdoerfer et al. (2017) argues that the type of relationship between these concepts differs among literature. In general, the connective nature between these terms can be more conditional (CE is a condition for sustainability), beneficial or trade-off oriented. Apart from differences, there are some important similarities that indicate CE is a relevant solution for sustainable development. The concepts share the challenge and commitment of solving environmental issues on a global scale, both decoupling the notion of development from economic growth. At the core of sustainability and CE is the idea of system change, emphasizing the necessity of cross-stakeholder cooperation, as well as implementation tools such as regulations and incentives. Overall, Nikolaou & Tsagarakis (2021) conclude from their literature review that CE is seen as an essential component to achieve sustainability goals. Contrariwise, Kirchherr et al. (2017) indicate that CE's tie to sustainable development is weak when looking at CE definitions.

CE in itself is a diffuse concept that has been evolving since the late 1970s, which until this day still has distinct meanings for different people (Geissdoerfer et al., 2017; Kirchherr et al., 2017). A frequently employed definition is formulated by the Ellen MacArthur Foundation, a renowned promoter of CE, defining the notion as *"an industrial economy that is restorative or regenerative by*

intention and design" (*Ellen MacArthur Foundation, p.14 2013;* Kirchherr et al., p. 226 2017). The CE highlights the cyclic value of materials and presents the possibility of a sustainable production-consumption culture within a sharing economy (Korhonen et al., 2018).

Over time, the idea of a closed loop has incorporated a variety of different features and components. More recently, Kirchherr et al. (p.229, 2017) formulated a CE definition by analyzing 114 definitions to acknowledge multiple aspects brought up in CE literature. Serving as a conceptual foundation, it reads:

"An economic system that replaces the 'end-of-life' concept with reducing, alternatively reusing, recycling and recovering materials in production/distribution and consumption processes. It operates at the micro level (products, companies, consumers), meso level (eco-industrial parks) and macro level (city, region, nation and beyond), with the aim to accomplish sustainable development, thus simultaneously creating environmental quality, economic prosperity and social equity, to the benefit of current and future generations. It is enabled by novel business models and responsible consumers."

The often under-addressed social aspect is included in this definition. Observe that the social dimension of CE is a key challenge for circular realization, as the CE requires a demanding socio-cultural shift (Calisto Friant et al., 2020) but still often overlooked in definitions (Kirchherr et al., 2017). Interestingly, consumers and business models are mentioned, giving the definition by Kirchherr et al. (2017) another dimension.

To establish a more coherent idea of CE for the industry, the International Organization for Standardization (2024) has developed a CE nomenclature standard, including definitions. There, CE is described as *"an economic system that uses a systemic approach to maintain a circular flow of resources, by recovering, retaining or adding to their value, while contributing to sustainable development"* (*Svenska Institutet För Standarder (a) p.1, 2024*, n.d.). Generally, there are multiple interpretations of CE, encompassing different dimensions of circularity depending on the context and timeframe of the notion. Note that the number of CE aspects considered can be very limited, such as connecting CE solely to recycling, making the concept more or less narrow depending on the definition (Kirchherr et al., 2017).

CE is argued to be beneficial for the social, environmental as well as the economic aspects of sustainable development (Bocken et al., 2016). In concrete terms, pollutive emissions are reduced, jobs are created, and economic growth is fostered at the same time. There's an opportunity to maximize the value of materials and products and move towards zero-waste reducing environmental impacts (*Ellen MacArthur Foundation, 2013 p.7.*; European Commission (b), n.d.). However, the inspiring and promising concept of CE has several limits and challenges (Calisto Friant et al., 2020; Kirchherr et al., 2017; Korhonen et al., 2018; Skene, 2018). Primarily, the concept's inconsistency in understanding and application creates an initial barrier. Moreover, an eminent challenge is the demanding shift required to transition to such a system and the following management of CE. Interorganizational and inter-sectional material and energy flows are another complex area within such a circular system, requiring capacities such as collaboration. Furthermore, Korhonen et al. (2017) point out that a circular solution does not necessarily secure a sustainable outcome. Difficulties with measuring the impact of CE is a present topic in literature and governmental institutions (Calisto Friant et al., 2020). It is central to ensure CE leads to positive impacts and excludes any hidden negative effects.

Multiple aspects of CE are criticized in line with its feasibility (Calisto Friant et al., 2020; Korhonen et al., 2018; Skene, 2018). The idea of CE per se is subject to laws of physics, limiting the practicability

of this circular system. For instance, materials cannot circulate endlessly as thermodynamics make complete recycling impossible. Skene (2018) argues that circularity must respect natural systems and boundaries, making a structured solution such as CE suboptimal attempting to control ecological processes (Skene, 2018). Material, energy and biodiversity relationships are complex and pivotal components of CE, challenging the achievability of a circular system. Notable is the limited exploration of alternative circular pathways, making CE the deeper investigated and, in turn, most popular solution for sustainable development today (Calisto Friant et al., 2020).

Promoted and endorsed by the EU, companies are strongly recommended to transition towards a CE to take responsibility for their impact on planetary boundaries. Despite seeing the concept and its implementation is still in its early stages, possibilities and barriers have been investigated. The CE shows a range of opportunities for businesses such as differentiation potential, improved customer relations and reduced environmental impact (Herrero-Luna et al., 2022).

The benefits a closed loop system promises do not exclude the challenges that arise with circularity. Foremostly, established companies seem to have challenges shifting to a circular system (Frishammar & Parida, 2019). Kirchherr et al. (2018) identify cultural barriers as the most critical difficulty for a transition towards CE. Consumer interests and awareness, along with hesitant company cultures, were the most notable topics. On the other hand, the technological hurdle remains less urgent (Kirchherr et al., 2017). Shifting to a CE seems to remain unpopular (Nowicki et al., 2023), indicating the variety of challenges.

3.3 Laws and Regulation Driving the Shift to Circularity

Legislations and regulations, both current and upcoming, are a central driver for companies to shift from a linear to a CBM (de Mattos & de Albuquerque, 2018; Herrero-Luna et al., 2022). In Europe, EU is the biggest organ supporting sustainable development, encouraging and determining sustainable practices for all Member States. The overarching strategy for this vision is the EGD, remodeling Europe into a resource-efficient and competitive economy while striving to become the first climate neutral continent (European Commission (a), 2021).

To reach sustainable resource use, the CEAP has been developed as a part of the EGD encompassing a set of measures for products along the entire life span (European Commission (b), n.d.). The CEAP looks at enhancing sustainability from the design of products to their consumption and end-of-life as well as preventing waste. Products and sectors that use the most resources are prioritized as major circularity possibilities are seen in correspondence. Note that the first CEAP in 2015 was endorsed before the EGD, adopted in 2019, and a new CEAP was introduced after the EGD in March 2020 (European Commission (c), n.d.).

The sustainable products initiative lies within the new CEAP and includes the ESPR. This measure's objective is to increase products' circularity and efficiency by looking at aspects such as durability, recyclability and energy performance, making them more environmentally friendly. The ESPR defines performance and information rules for almost all categories of physical goods, named "eco-design requirements". These requirements are for example enhancing product maintenance possibilities, increasing recycled content and limiting products' environmental footprint. Observe that historically the Eco-design Directive 2009/125/EC is the antecedent of the ESP regulation (European Commission (d), n.d.).

In March 2022, the so-called DPP was introduced into the ESPR. This passport serves as a digital identity card for products, components and materials, storing relevant information to support product circularity. Such information can include materials, repair activities and environmental

impact. The idea is that information will be accessible electronically for simplicity and convenience of use in a big European database for relevant stakeholders to access (European Commission (d), n.d.). In June 2024 the ESPR regulation entered into force, including the DPP initiative. What information is yet to be decided by the European Commission and will depend on the type of product. The first group of products having to comply with the ESPR's law was set to April 2025 and the DPP decisions for the first category product category, textiles, is appointed to January 2026 (Circularise, 2023). An overview of the European strategies and regulations can be seen in Figure 3.2.



Figure 3.2: An overview of the European strategies and regulations.

Besides laws and regulations, there are other enabling factors for circularity. de Mattos and de Albuquerque (2018) believe education to be one of them. Information enables, for example, customers to make better purchasing decisions, consumption choices and optimize products' lifetime (European Commission (e), n.d.). Accordingly, the EU has implemented directives to help consumers in this green transition and protect them from greenwashing. It is believed that consumers are an essential factor for achieving CE as current consumption, where user behavior is central, needs to shift for circularity. Interestingly, there are studies showing a gap between the willingness of customers to engage in circular practices and their actual engagement. This could be due to a lack of product information, for example repairability instructions, and the absence of markets such as secondhand products and leasing services. Better or increased information is believed to help close this gap and assist consumers in becoming more circular (European Commission (e), n.d.).

The threat of new competitors, market demand and social pressures are other drivers encouraging companies to become more sustainable (Herrero-Luna et al., 2022). Furthermore, businesses are supported to shift to a CE through their own company culture by having interest in corporate social responsibility and inherent motivation to change current business practices into circular practices (de Mattos & de Albuquerque, 2018; Herrero-Luna et al., 2022). A combination of all factors mentioned may be needed for optimal motivation for companies to become sustainable (Herrero-Luna et al., 2022).

3.4 Strategies For a Circular Economy

Today, there are no specific CE application guidelines and circularity is rather seen as a mindset for companies (Frishammar & Parida, 2019). However, multiple strategies, principles and frameworks have been developed to aid the challenging adoption of this new circular system (Bocken et al., 2016; *Ellen MacArthur Foundation, 2013*; Kalmykova et al., 2018).

Overall, circular approaches are developed based on the CE principles - eliminate waste and optimize resource use throughout the lifecycle of products and materials as well as regenerating nature (Circularity Deck, 2021; Ellen MacArthur Foundation, n.d.). More particularly, circular strategies are built on so-called resource cycles, the process in which resources flow. Deriving from these, there are essentially two main cycles on which sustainable methods are built: goods can be reused or goods can be recycled. These methods can be further categorized into slowing and closing looping strategies (Bocken et al., 2016). Apart from these two actions, sustainable development can be further supported by using a narrowing or regenerative approach. These strategies imply decreasing resources as well as using renewable materials and energy (Bocken et al., 2016; Circularity Deck, 2021). Additionally, information is a method believed to support these four key strategies attempting to shape energy and material flows (Circularity Deck, 2021).

A different way to categorize circular strategies is, for example, by creating a database. The idea is to give guidance and provide a search engine on existing strategies to help companies implement CE (Kalmykova et al., 2018). Kalmykova et al. (2018) classify the specific strategies according to life cycle phase, giving an overview of what can be done at each stage of a product's lifespan, compared to adopting a wider slowing, closing, narrowing or regenerative strategy at different levels.

Circular strategies can be implemented on multiple levels - the product, business model and ecosystem level (Circularity Deck, 2021). Bocken et al., 2016 argues that there are two main ways to approach circular operations - design and business model strategies. The first being a rather proactive action, interfering in the design phase of products and services to enable a loop, and the latter showing a more holistic perspective looking at the offerings of a business as a whole and what changes are required for circularity in multiple aspects. More specifically, circular design strategies on a product level focus for example on designing for long-life products, product life extension and design for dis- and reassembly. It is believed important to integrate circularity in the early stages of products to ensure they are designed to have value across their entire life cycles (Bocken et al., 2016). Note that within circular design practices there are multiple Design for Sustainability approaches that have been created over time to solve different sustainability challenges on levels ranging from product to the socio-technical system stage (Ceschin & Gaziulusoy, 2016). Distinctively, the business model level strategy integrates sustainable practices at the core of how companies do business (Bocken et al., 2016). These models include different ideas on value creation and resource management. Replacing physical products ownership by providing their capability and performance as a service, often called Product Service Systems, is one example of a CBM strategy. Other business approaches include long product life models, offering a durable design and repairability possibilities, and extending resource value through collection and sourcing of "wasted" resources (Bocken et al., 2016). Such business models provide the opportunity to create competitive advantages, staying profitable, while addressing sustainability challenges.

Bocken et al. (2016) argue that design and business model strategies need to be implemented in combination with each other as these strategies go hand in hand and address different aspects of resource management. The ecosystem perspective is believed essential for a working CE (Circularity deck, 2021).

In practice, R-strategies - encompassing the principles of reduce, reuse and recycle - support circular design and business models by guiding how resources can be reused and waste can be kept out of the loop (Samani, 2025). These practical principles give more hands-on advice to assist companies in their circularity adoption but also help questioning the current idea of resources and consumption patterns (Circularise, 2023). There are even more advanced frameworks, such as the widely accepted 10R strategy, incorporating additional methods acknowledging the entire product life cycle (Samani,

2025). See Table 3.1 (Circularise, 2023). Note that multiple strategies are often implemented in the same project (Circularise, 2023).

Table 3.1: An overview of the 10R-strategies

Strategy	Description
R0 Refuse	Prevent the use of products and raw materials in the creation of goods, processes and services.
R1 Rethink	Reconsider ownership, use and maintenance of products.
R2 Reduce	Decrease the use of raw materials in products and services.
R3 Reuse	Secondary use of products by another owner for the same intended purpose.
R4 Repair	Maintain and repair existing products for extended use.
R5 Refurbish	Restore and improve products to a satisfactory condition for extended use.
R6 Remanufacture	Make more products with the same purpose with discarded products or parts.
R7 Repurpose	Make new products with a different purpose with discarded products or parts.
R8 Recycle	Process waste into new products or materials that can be used for new products.
R9 Recover	Process waste to recover energy.

Transitioning to become a sustainable company is challenging as there are no step-by-step guidelines for such (Frishammar and Parida, 2019). Subsequently, the International Standards Organization (2024a) has developed a standard to aid the implementation of CE. To define a strategy, in short, companies are strongly advised to define sustainability and develop specific sustainability goals as a first step. As a second step, challenges and opportunities are identified considering circular actions. This creates the building block for the circular strategy of the company. Observe that the standard underscores the importance of integrating circular principles into organizational practices when introducing CE. Once a strategy has been developed, it is implemented and evaluated (*Svenska Institutet För Standarder (b), 2024*).

Implementing and, especially, assessing the performance of a company's sustainable development strategy is a further challenge that needs methods when transitioning towards a CE. Measuring circularity is complex as the concept of CE is ambiguous, encompassing diverse understandings, but also contains various elements (Moraga et al., 2019). Today, most advanced indicators focus on resource preservation, such as recycling (Moraga et al., 2019), and the metrics fail to address all circularity dimensions as their focus lies on environmental and economic aspects (Corona et al., 2019; De Pascale et al., 2021). Additionally, there is a limited connection between circularity indicators and environmental performance (Harris et al., 2021) which enhances, for example, the risk of shifting the problem from reduced material consumption to increased environmental impact (Moraga et al., 2019). It is believed that either multiple indicators are necessary to effectively measure progress towards CE (Moraga et al., 2019), ideally, a simple but absolute model containing all necessary aspects of CE for measuring circularity should be developed (De Pascale et al., 2021). Interestingly, De Pascale et al. (2021) argue that data availability and quality are necessary for good measurements. Data can contribute to multiple indicators and help acquire representative results (De Pascale et al., 2021). While various measuring methodologies and indicators exist, the effectiveness of CE strategies is still under inspection and yet to be determined.

3.5 Incumbents Transition Towards Circularity

Incumbent firms, organizations already established in the industry, play a critical role in the advancing field of sustainable development. These firms can both enable sustainable development, as well as hinder it. On one hand, their size and access to complementary assets, such as a wide network of suppliers, established distribution channels and service infrastructure, create potential for large-scale impact. Their involvement can accelerate sustainable transitions and environmental innovations through effective usage of their complementary resources (Geels, 2011). On the other hand, a transition towards sustainability and CE will often disrupt these existing structures, relationships and potentially the offering to customers, presenting obstacles to change (Frishammar & Parida, 2019).

One strategy to overcome these limitations and facilitate the transition towards circularity is the adaptation of CBMs. Such business models incorporate CE principles into the firm, focusing on improving for example resource efficiency and extending product life-spans (Frishammar & Parida, 2019). CBM strategies often focus on slowing and closing resource loops, some examples are product-as-a-service systems, product life extension by designing for durability and repair, and exploring second-life opportunities for waste. CBM implementation and circular design strategies should be implemented simultaneously to succeed. For incumbents, CBMs can provide a structured pathway for transitioning towards circularity. However, changing established business models has proven difficult. Business model innovation often requires substantial investments, creating a trade-off between short-term profitability and long-term sustainability goals (Bocken et al., 2016).

Although a transition towards CE can be challenging for incumbents, there are significant benefits and opportunities that arise with such a shift. Engaging in social and environmental practices can provide companies with both short-term value and organizational resilience. Organizations who are sustainability oriented have also been shown to be more resilient to changes in their ecosystem by adjusting quicker. They develop a better ability to sense their surroundings and anticipate changes, making them better prepared for a shifting environment (Ortiz-de-Mandojana & Bansal, 2016).

As incumbent firms adopt CE principles, the success of the transition can be influenced by their organizational structure. Different organizational structures can either enable or constrain the transition, depending on how easily changes in operations can be implemented throughout the entire organization (Pascucci et al., 2024). One key distinction is the one between centralized and decentralized decision-making structures, each creating implications for implementation of circular strategies. A centralized structure facilitates communication between units or regions, making common strategic decisions, business practices and procedures easier to implement across the entire organization. On the other hand, a decentralized structure allows for quick adaptation to local conditions. By adopting a “learning by doing” approach, multiple strategies can be tested at once at different locations. Commonly, organizations tend to use a hybrid approach for decision-making to reap the benefits of both structures. A centralized approach is used for overarching circularity targets, while the decentralized approach allows for agility and innovation locally (Fagerlind et al., 2019).

When transitioning towards circularity, implementing new processes and ways of working into the organization to realize new initiatives is a common aftereffect. In the case of incumbent firms, they tend to tackle new problems and initiatives by leaning on their established routines, existing knowledge and assets. In this way, incumbents are often path-dependent, relying on the success from past strategies and work processes (Teece, 2007). At the same time, a necessity for success when introducing circular practices in organizations is to integrate them into operations and processes, meaning that established routines and ways of working might need to shift. It is a strategic assignment

requiring involvement of managerial instruments to support the process of implementing sustainable activities, all the way from planning to review and control (Baumgartner, 2014).

A multi-level management model developed by Baumgartner (2014) reports how to integrate sustainability into the corporate strategy through alignment of normative values, strategic planning and operational activities. Figure 3.3 shows three types of management levels and their role in implementing sustainability actions into a corporation.

Normative Management – Objective Legitimation
Determine the basic management philosophy
Values, attitudes, beliefs, and judgments as a basis for management
'Who are we and who do we want to be?'
Strategic Management – Objective Effectiveness
Determine long-term goals and product/service-market combinations
'What are our (strategic) goals?'
Operational Management – Objective Efficiency
Organization and management of all corporate activities to reach strategic goals and guidelines
'How can we reach our goals?'

Figure 3.3: Types of management levels

Baumgartner (2014) states that normative management encompasses the elements of stating and investigating the corporations' core values, vision, mission and organizational culture. A common vision is essential for successfully implementing sustainable actions. Additionally, successful implementation relies heavily on organizational culture, incorporating employee attitudes and openness to sustainable development. Organizational culture furthermore incorporates creating shared values and norms which align with the vision and mission stated by the highest managerial instance.

Strategic management includes planning and translating normative visions into actionable strategies. This stage aims to provide a roadmap for how the vision and mission of the corporation will be translated and fit into the specific operations of the company in question. Doing this requires investigation of the external environment as well as internal strengths and weaknesses. The strategic planning of sustainability strategies can be done through forecasting, meaning to look and assess external developments and foresee future ones, or through backcasting, meaning to first establish a future goal and establishing strategies and actions afterwards. For sustainability problems, Baumgartner (2014) states that backcasting is the most suitable approach.

On operational management level, the strategy is implemented and executed practically in the business' daily functions. Each company functions specified sustainability-oriented activities should be defined and carried out. Figure 3.4 illustrates the recommended phases of corporate sustainability management by Baumgartner (2014). It is furthermore recommended to include feedback loops between the phases for the best result.

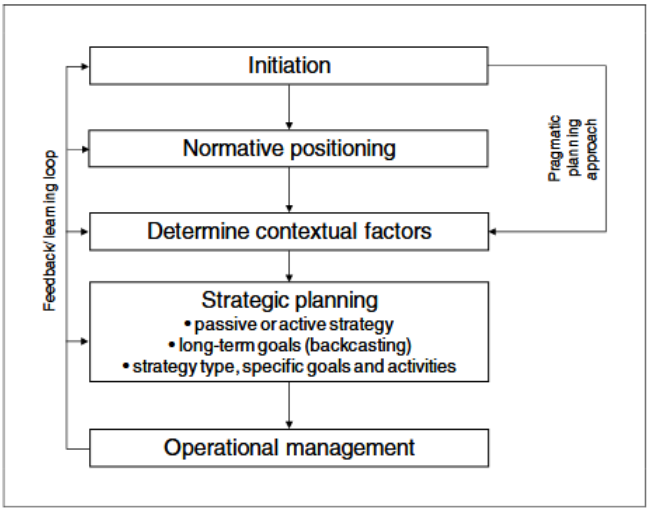


Figure 3.4: Phases of corporate sustainability management.

3.6 The Role of Product Life Cycle Information for Circularity

Information has been identified as an essential enabler to foster CE. Correspondingly, it has been argued that information flow is just as important to consider as material flow to permit the principle of circular products and materials (Jäger-Roschko & Petersen, 2022). To prove this, studies have shown that information streams have a positive impact on CE (Jørgensen & Scarso, 2023).

Information needed to perform circular practices depends on the nature of that activity (Jørgensen and Scarso, 2023). Provided data and effective information exchange permits sustainable improvements on different levels for multiple stakeholders (Jäger-Roschko and Petersen, 2022). To name a few, data, for example in the form of feedback, allows design and manufacturing process improvements for companies. In turn, adapted design permits product disassembly and enhanced material recyclability. Additionally, information enables faster and more predictable maintenance, saving costs for the consumer and creating profit opportunities for businesses (Jäger-Roschko & Petersen, 2022). On a consumer level, knowledge sharing is expected to support proper product use, facilitating product maintenance and repair as well as correct product disposal (Jäger-Roschko and Petersen, 2022; Jørgensen and Scarso, 2023).

Even though information flows have shown promising opportunities for multiple stakeholders and advancing CE, there are various barriers across the product life cycle to such implementation. Firstly, acquisition of data is complex. Even if information exits, data must come from multiple sources such as manufacturers (Saavedra-Rubio et al., 2022). Brought up in the theme of Life Cycle Assessments, gathered information may also be inaccurate and there is a lack of verification mechanisms for LCA results (Sala et al., 2021). Furthermore, companies might be reluctant to share information as certain data can be classified as confidential (Harkonen et al., 2024). Balancing transparency and secrecy are challenging as data can be valuable and used for competitive advantage (Harkonen et al., 2024). Particularly, challenging is the usage phase (Adisorn et al., 2021). Motivation is needed to increase the willingness to provide information and ensure the completeness of product data for all its life cycle phases (Wan & Jiang, 2025). Overall, information asymmetries and inconsistencies, due to incomplete data, is the initial barrier for information flow (Harkonen et al., 2024).

When it comes to technological barriers, information flow requires the linkage of multiple IT systems. Managing information can be costly and may require companies to invest in technological infrastructure and personnel (Harkonen et al., 2024). Furthermore, the costly nature of data collection may result in a lack of motivation for companies to adopt this CE strategy (Jensen et al., 2023). Economic priorities, along with technological challenges, make the implementation of the already complex product life cycle information collection and traceability demanding.

Life Cycle Assessments use data to measure environmental impact of products, which is important to understand the impact of products and potential circular practices (European Environment Agency, n.d.). In LCA's, the Life Cycle Inventory (LCI) phase, where data collection is performed, is also described as a resource-intensive and time-consuming step for companies. However, it is an important and crucial step as data is the backbone for a reliable LCA study. Saavedra-Rubio et al. (2022) also mention the lack of detailed procedures for the complex step of data collection. Furthermore, Sala et al. (2021) show that there is a lack of standard LCI rules across the EU, meaning that different companies may follow different protocols. Additionally, as LCA's require extensive amounts of various data, the lack of trained personnel to understand and use data properly is brought up in connection with performing LCA's. Companies turn to for example consultancies to overcome lack of trained personnel (Gómez-Garza et al., 2024). While gathering data is one challenge, the amount of already gathered information should also be considered and be taken advantage of (Kuittinen & Häkkinen, 2020). The lack of detailed procedures and trained personnel makes life cycle information gathering, as well as understanding and using that data correctly, a challenge for performing LCA's, in turn affecting circularity.

The implementation of the upcoming DPP regulation underscores the key role of information flow and sharing for circularity. The idea is to have product information gathered by multiple stakeholders and stored digitally to improve circular practices and help users make better purchasing decisions (Lopes & Barata, 2024). However, there is no agreed format for DPPs today as the passport is still under development in EU institutions. Types of information to be collected and available technologies to support DPP have been explored to attempt to tackle the challenging transdisciplinary task of information management (Lopes & Barata, 2024). To give an idea of what and how DPP will work in practice, the review by Lopes & Barata (2024) shows possible DPP information types to be tracked, see Table 3.2.

Table 3.2: Most common structural elements of a DPP.

Information	Details and examples
Product information	Materials sourcing data (Date and Place); Compositional elements (materials, substances, components & parts); Properties (physical and chemical properties, material criticality, bio sustainability, recycled content, reused/refurbished parts, hazardous content to human health or the environment)
Environmental information	Environmental footprint; Carbon footprint; Reusing, Reprocessing, Recycling Data; Information about the product in terms of sustainability and circularity properties (e.g., environmental impact, social impact, circularity performances, disassembly instructions, inventory data used for calculation, calculation methods, indicators for environmental/social/circularity performance, applied standards for impact assessment, product design)
Manufacturing Data	Manufacturing process and remanufacturing, Resource consumption/Consumption in production (water, energy, etc.)
Utilization Information	Documentation of any parts of a product that have been replaced or repaired, maintenance history, delivered performance
Life cycle Data	Expected percentages of products that can be recycled and the waste part; Documentation on collection, sorting, and treatment during the end-of-life of a product; Recovery Data
Supply chain	Suppliers of components; Recycling partners; Customer return channels; Implementation of CE business models; Information about value chain actors involved
Guidelines and Manuals	Product manuals; Waste sorting guidelines; Safety instructions
Compliance	Regulatory/voluntary compliance; Product-specific regulation; Labor conditions

Popular technologies, according to literature, identified by Lopes & Barata (2024) are presented in Figure 3.5.

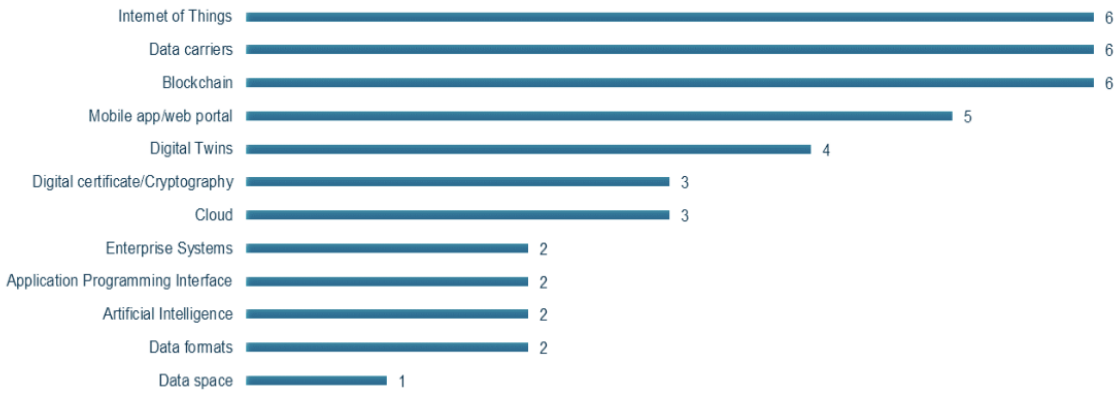


Figure 3.5: Most popular DPP Technologies.

Each technology has its advantages and limitations. Harkonen et al. (2024) argue that it is essential to understand the strengths and weaknesses of different technologies to implement the most suitable

technology. Possibly, multiple technologies could be combined to meet the traceability demands for the specific product. A traceability technology can be used for internal purposes of the company along with other technologies for external actors to access (Harkonen et al., 2024). It is expected that the types of information required would be tailored to category of products (European Commission (d), n.d.). Additionally, each product will need a data carrier sustaining DPP information, meaning that technologies need to be considered carefully, and systems possessing this data need to be interoperable with other systems (Circularise, 2024). Guidelines for technologies and databases are also said to be decided and proposed by the European institutions (European Commision (f), n.d.).

Overall, end-to-end traceability, which is supported by the complex action of information gathering and flow, enables CE initiatives such as reducing waste and conserve resources. This is a result of reuse, refurbishment and recycling actions unlocked through product life cycle information (Harkonen et al., 2024). More proactively, information provides product related insights that can inform decision making at different life cycle stages (Walden et al., 2021).

4 INTERVIEW RESULTS AND DATA ANALYSIS

In this chapter, circularity practices and life cycle information gathering at the case company are investigated. The material in this section originates from interviews that the authors held with internal and external stakeholders connected to the case company. The process of attaining interview material and its analysis is further specified in sections 2.4.2 and 2.5.1. The authors utilized 15 codes, derived from the research questions, to analyze and synthesize the attained interview data. These codes were further divided into 4 bigger themes to present the information in a logical way. These are presented in Figure 4.1.

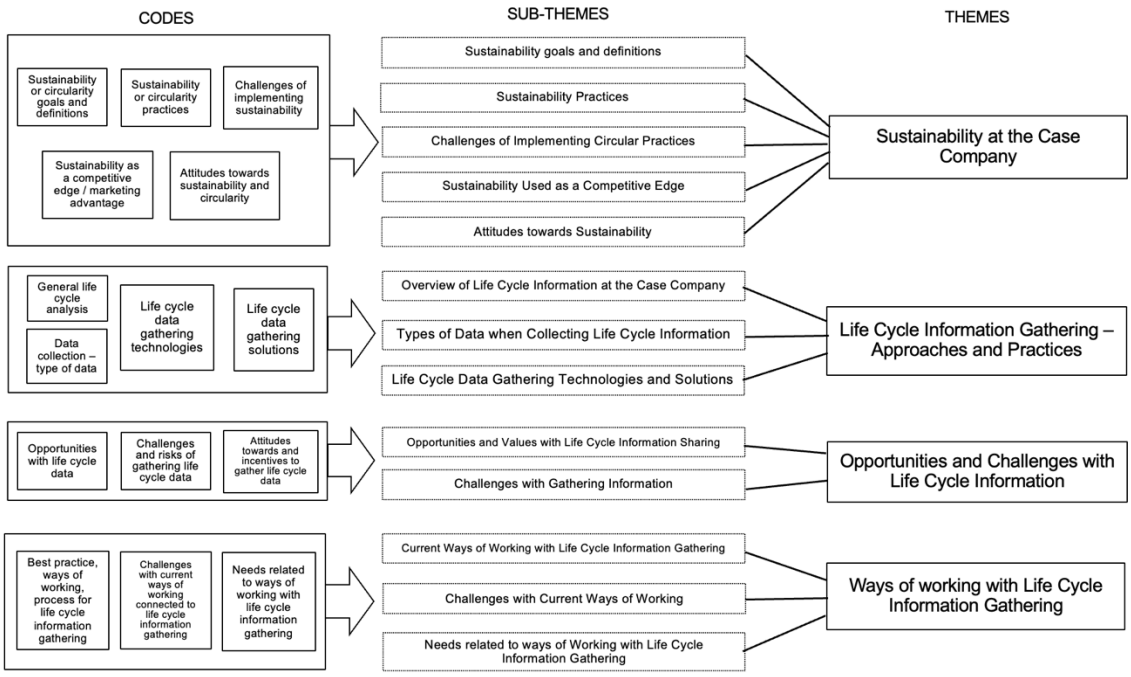


Figure 4.1: The codes and their respective themes.

Accordingly, this chapter is divided into the four main themes: Sustainability at Case Company, Life cycle Information Gathering – Approaches and Practices, Opportunities and Challenges with Life Cycle Information Gathering and Ways of Working at The Case Company Including Life Cycle Information Gathering. Under each of the themes, the relevant codes are disclosed and analyzed.

4.1 Sustainability at Case Company

This section aims to provide an overview of current sustainability and circular practices at the case company. Although not directly connected to a research question, data provided in this section is seen as important to understand the result.

4.1.1 Sustainability Goals and Definitions

Respondents across all internal interviews agreed that sustainability has become a more central and important question for the case company in recent years.

"Before you've had sustainability in the back of your head, but maybe you haven't worked so intensely with it. But really now it's the entire game plan for us to think sustainability all the way" - Respondent

Respondents also agree that an increasing number of their customers are requesting sustainability. Examples brought up by respondents are that customers are more interested in attaining information on the purchased products' environmental footprint, conditions at the case company's suppliers and material mixes of products.

The fact is that sustainability work is present across all interviewed departments in different ways. At an Original Equipment Manufacturing (OEM) production site, a goal of limiting combustible waste to a maximum of 15% has prompted them to collaborate with external stakeholders who utilized scrap-bits of hoses in other applications, for example in concrete production and asphalt. The collaborations were a success, and combustible waste was kept below 15%. However, it was also revealed how sensitive these external collaborations can be. For example, in 2024 the usage of hoses in asphalt was prohibited due to environmental restrictions from the EU, which caused combustible waste to rise above the set target. This shows the importance of being proactive in sustainability work and keeping track of how external stakeholder decisions will influence the company's own environmental impact. Such external regulations were also present in the work with material compliance, where a respondent explained the importance of keeping track of materials that will be phased out or prohibited to use in products. The customer of the case company utilizes the products in another application, for example a truck or excavator, which means that the material in the products need to comply with restrictions in other industries than solely the "hose industry". Once again, the importance of a proactive way of working was brought up by respondents. This is essential to keep track of upcoming restrictions that concern products and updating the product assortment accordingly.

When the respondents were asked about what the case company's sustainability definition and goals are, there was no clear answer among respondents. The case company's sustainability definition, vision and strategy are not known amongst most of the respondents. What was often mentioned, however, is the case company sustainability report and the EcoVadis ranking of the company. Although, there is a common agreement that sustainability is something that should be present across the company's departments. Some respondents speak about sustainability in a broad perspective, considering the whole life cycle of the products, others speak about it in terms of what can be done within the borders of their department.

One respondent reflected deeper on that it can be difficult for the case company to pinpoint where their sustainability actions shall be directed. As the case company products are utilized as a part of another application, the products themselves are not directly connected to negative environmental impact, for example emissions. In this sense, the company is not the direct source of sustainability issues, except waste generation, making the linkage to circular practices more abstract. It is believed by the respondent that there are no "obvious" or "apparent" vision or goal for the case company in terms of sustainability.

4.1.2 Sustainability Practices

Although respondents feel that the case company has not set a clear definition of what sustainability entails for them, there are a few common sustainability and circularity enabling practices present across the organization. One example brought up by most respondents is the work with EcoVadis, an independent platform which ranks companies according to how they work with sustainability questions. The platform helps companies identify where their strengths and weaknesses lie within

sustainability work. A company can attain four different rankings: bronze, silver, gold and platinum. The case company recently attained certification for the silver level, which is something the respondents mention frequently and are proud of. The respondents feel that these kinds of certifications drive the sustainability work forward and are an effective argument to use towards customers interested in the case company's sustainability work. Other certifications mentioned by respondents is the ISO 14001:2015, the internationally recognized standard for environmental management systems.

There is a vision to consider environmental sustainability at the case company throughout the product's life cycle, ranging from supplier contact to installation of product at the customer. Respondents mention that more knowledge on products usage could for example allow them to make better lifetime estimates and design improvements for more durable products.

Regarding supplier contact, respondents bring up the “Supplier Code-of-Conduct”, a document which has to be signed by all suppliers through the case company SRM tool to ensure they comply with the case company's values and vision regarding social and environmental sustainability. At larger suppliers, both self-assessments performed by the supplier themselves, and on-site audits performed by the case company is an important tool to identify shortcomings in social and environmental sustainability practices.

Regarding transport from supplier to warehouse, the respondent mentioned that there have been efforts to look into more environmentally friendly means of transportation. For example, switching from trucks to trains. However, the respondents mention that such efforts are often dismissed due to cost or reliability.

When it comes to product development, the case company is limited in their ability to change the properties of products, for example material composition or manufacturing methods. The case company purchases ready-made product parts from suppliers and assembles them, which means they have limited abilities to influence the contents of the product's parts. However, respondents report that the case company make efforts to find multiple application areas for each of their product article, with the aim of decreasing the number of articles in their product catalogue. For example, a fitting designed for a hydraulic hose might also be useful for a low-pressure hose if some small changes are made. By doing this, they can keep their stock moving and avoid keeping products that decrease in quality over time in stock, reducing waste.

Respondents report that when a product is sold at a store, the customer pays an environmental fee. Paying this fee allows the customer to bring a used hose, that has broken or needs replacing, back to the store upon purchase of a replacement. The fee covers the waste handling cost for the case company. Wasting old hoses, especially the ones which have been in contact with hydraulic oil, is costly as it requires the waste management company to handle it as hazardous waste. Respondents report that this fee allows the case company to handle the waste in a way that is compliant with laws and reduces negative environmental impact.

Additionally, respondents report that employees at stores aim to keep sustainability in mind in the day-to-day operations. One example is that they take note of returning customers who bought a product that has broken unusually fast and try to investigate why this might have happened. Wrongful installation or handling of the product can cause it to break faster than intended. In these cases, store employees advice the customer and attempt to find a product that can last longer in their specific application, reducing waste. Store employees are urged to minimize waste by saving and utilizing scrap bits of hoses, for example whenever a customer requests a shorter hose.

4.1.3 Challenges of Implementing Sustainable Practices

Although the case company has implemented several practices which reduce their environmental impact, respondents discuss several challenges connected to sustainability work.

Firstly, the fact that the case company operates in several geographical locations adds complexity in creating common protocols that are applicable everywhere. Environmental laws and regulations can vary a lot depending on the global geographical location, making it necessary for the case company to adjust their sustainability goals to the local context. Respondents note that this makes it challenging to create a common way of tackling sustainability, as there will always be exceptional cases that require another approach. Due to this, the case company operates in a decentralized manner, meaning that all stores, markets and countries have a high degree of individual decision-making authority. While respondents mention this is necessary in order to adapt to local needs and requests, it adds to the complexity of implementing common sustainable ways of working.

"Okay, if we are going to talk about sustainability, we need to find a way to do it in a decentralized manner without losing the core of what we want to achieve." - Respondent

Respondents feel that while the decentralized organizational structure allows them to be customer centered and adapt faster to local needs, there has to be some common directives regarding sustainability work set if the case company wants to progress in their journey towards sustainability.

The case company prides itself on its availability of products and to anticipate and meet customer requirements. In this sense, they are a very customer centered business. Respondents across all interviews mention that even though this is part of what has made the company so successful in comparison to competitors, it can complicate sustainability efforts. For example, it can lead to a conflict between sustainability targets and inventory levels. Even though the great availability of products creates a competitive edge for the case company, respondents mention that the balance between always providing the customer with what they want and working in a more sustainable way is a great challenge for the case company.

4.1.4 Sustainability Used As a Competitive Edge

In 2023, the case company released their first sustainability report, which respondents agree on has been a determining step in the journey towards sustainable practices. Apart from providing employees at the case company with information regarding the sustainability situation, respondents mention that it attracts customers. Some customers of the case company value sustainability work highly when determining their supplier. Respondents working within sales mentions that in these cases, the sustainability report has been an important tool to utilize.

"For sales the sustainability report is used as a tool, it's regarded to be very important tool today. It opens a lot of doors, which is very good" - Respondent

Sustainability has become increasingly important for the case companys' customers, especially larger industrial ones. Some of the largest customers even demand that the case company is able to show proof of their sustainability work before they buy their products. The type of information requested by these types of customers ranges from how well the case company knows their suppliers, to what their current sustainability initiatives are. Non-industrial customers, purchasing products at a store for example, tend to be less influenced by how sustainable the products they purchase are, according to respondents. Although, respondents have noted that these types of customers can be interested in how waste is handled by the case company.

4.1.5 Attitudes Towards Sustainability

All respondents agree that the case company has made significant improvements in working with sustainability in recent years. Today, the attitudes towards working with sustainability have been noticeably improved compared to previously. Respondents have noticed how new directives and guidelines from the top management incite employees to apply more sustainable ways of working in their respective departments. Respondents note that employees at the case company appreciate being included in the new sustainability efforts.

"We have changed our way of working in order to engage everyone so they will think more about that part, it becomes more of a common effort from the whole team and not just one person responsible for these questions" - Respondent

For the case companies' customers, respondents have seen mixed reactions when sustainability is mentioned. While large industrial customers value or even require sustainability, non-industrial customers can be more hesitant. One example brought up by a respondent is when an employee suggests adding a tracking label to a hose, the customer can be hesitant if they have not used this before. In these cases, it is important for the employee to explain how the customer can benefit from adding this kind of tracking label to their product, and how it helps the company be more sustainable.

4.2 Life Cycle Information Gathering - Approaches and Practices

This section provides a general overview on what life cycle information is and how life cycle information gathering is done currently at the case company.

4.2.1 Overview of life cycle information at the case company

Collecting and tracking product life cycle information across products' entire life spans is not something that is performed throughout all instances of the case company today. Nevertheless, certain product information is gathered in different life cycle phases using various systems. Most of these systems are not connected today and do not communicate with each other, in other words, do not exchange or transfer data. Respondents also specified that life cycle information is often reported in external stakeholder systems, not connected to the case company platforms. Types of information collected, technologies and the systems mentioned are further discussed in the upcoming sections 4.2.2 and 4.2.3.

When the products move internally, when received from the supplier before it is handed over to the customer, limited information is collected and forwarded to the sales point. The first challenge is that not all information about products is given from suppliers, such as precise material composition, due to company secrets. Additionally, the company does not actively request all information, besides safety and compliance related data. However, certain customers request product specific information to be transferred into their own system.

In the purchase phase, customers receive the opportunity to track their product order using the company's own tracking label. Respondents mention that attitudes towards tracking life cycle information at that stage vary substantially depending on the stores' mindset and customers' demand. Interestingly, even though demand varies significantly, those tracking their product only report positive feedback about traceability. This is often associated with experiences revealing the benefits of data collection.

"We only receive positive feedback" – Respondent

Respondents at case company report at multiple instances that what they offer to their customers is more than just a hose, it is a service for solving the customer's problems. Regarding this company offering, a recurring problem today is the absence of information about hoses' utilization, such as precise information about the hose's position on a machine, operating time or fluid. This poses challenges when figuring out performance errors and providing customers the best solution.

Respondents have mentioned prior discussions about making it more of a standard to track multiple types of information across life cycle phases, especially as the products leave the case company. This can help service technicians and also provide better service to the customer. It is mentioned that a significant driver to initiate such are competitors.

"The vision is that all products that we sell have a tracking label or serial number that we can follow up." - Respondent

The respondents reveal the connection between product life cycle data and circularity by illustrating different examples where information acts as an enabler for circular practices. More precisely, it has been mentioned that lacking product information, such as which fluid the product has contained, has prevented exploration of circular initiatives. For example, exploring possible recycling opportunities of hoses that today are classified as hazardous waste has not been possible as there is no data on which fluid it has been in contact with.

4.2.2 Types of data when collecting life cycle information

Respondents state multiple types of data when referring to product life cycle information. These types of data are, or are sought to be, collected and recorded in different phases of the products' life spans. Product life cycle information is mentioned both regarding data collected today and wished data to track.

When products are bought from suppliers, there are always physical marks on hoses displaying data such as type of hose, manufacturing year, brand and article number. Moreover, respondents mention that further data, in the form of security data sheets and its summarized version called safety data sheet, is usually received in addition to the physical imprint. These documents inform about risks in relation to the product and its usage. Interviewees also point out that suppliers have their own product catalogue with technical information and claims. Information beyond these, such as specific material composition is rarely obtained. Today, the case company has the possibility to figure out material composition and verify other performance data through in-house expertise and their testing facility.

Internally, the case company uses both an Enterprise Resource Planning (ERP) and a Product Information Management (PIM) system to log specific product information for all products in their assortment. The ERP system registers "basic" product information including weight, hose "expiration date", life cycle status (whether the product can be sold or not), which assortment (local or group) and sales quantity (sold in meters for example). Here the products are created and further transferred to the PIM system. In the latter system, technical data, working pressure, hose measurements (inside and outside diameter), eligible temperature and material data are logged. The interviewee indicates that the systems only communicate in one way, from ERP to PIM, meaning that changes in the PIM system must be transferred manually to the ERP system.

At purchase*, the customer gets the opportunity to collect information about their order using a physical tracking label on their product. The customer's unique article number, the company's internal serial number and the purchase date are added onto the label. Additionally, a personalized label name can be added if needed by the customer. The internal serial number permits tracking of the order's elements which can be checked at all times in the case company's ERP system. Annotations can also be added in that system, but the label does not trace more information than that. Apart from the pickable tracking label, information in the form of installation examples can be found in the company's product book to aid optimal set up.

The mobile hose service working with emergency replacements and preventative maintenance can also provide the tracking label to customers. At assignments, the mobile hose service reports information in terms of a risk analysis checklist for safety and environment, service reports and service protocols. These documents consist of data like travel information, used products and other annotations deemed necessary by the service technicians.

Respondents show interest and desire in tracking more product life cycle information than what is done today. Overall, respondents think information should be asked more actively and profoundly as it is believed to help understand potential performance issues or simply to assist in providing the best service. These questions can for example be about the hose's installation, operated pressure numbers and flow type used, as these kinds of factors affect hose lifetime drastically.

"If someone enters [the store] with a hose it should, for example, always be asked: Okay how long has it been installed?" - Respondent

Multiple interviewees express that ideally extensive product life cycle information should be collected and tracked. However, it is important that the amount of data tracked is feasible for the employee.

"In the best of worlds you would want all types of information accessible" - Respondent

Overall, when it comes to what type of data, it should be reported how gathering of it is done, where it is done and when it is done. These questions can be translated into a range of data points to collect. Among others, respondents have mentioned the following data to be interesting: when the hose has been assembled, when it has been ordered and exchanged, how long the previous hose held, how the hose is installed, where precisely it is placed, what kind of media flows through it and when the next inspection is planned if maintenance is planned. Additional information such as pictures, documents, checklists and additional comments should preferably also be gathered in some way.

4.2.3 Life cycle data gathering technologies and solutions

Respondents reveal that different solutions, supported by various technologies, are used to collect and track product life cycle data. Solutions used depend on the phase of the product's life span but also substantially on the requests of the customer and industry standards. Both internal and external solutions that are used are discussed by respondents.

A specific communication platform has been integrated recently to facilitate information sharing between suppliers and the case company. On this platform, the supplier code of conduct is approved and self-assessments are sent out to suppliers. This platform was established to gather information connected to suppliers in one place but also for easier contact with suppliers, as communication

* Customers can purchase products from the case company's stores, OEM production sites as well as directly from their storage facilities.

degree with suppliers varies significantly across the landscape. This system also gives the possibility to gather more information in one place.

Customers sometimes request material data from suppliers. In that event, the case company adapts to the client and reports the material to the desired system. For example, the case company uses the International Material Data System (IMDS), a system free of charge created by the automotive industry, to collect material data from suppliers. Another system, called BOMcheck (build of materials), is also used sometimes to collect and report data. All these systems are mostly used as a tool to aid compliance with substances regulations. The respondent notes that the user friendliness of different material reporting systems vary considerably. No specific internal tool, except for the ERP and PIM system, is used to collect life cycle information from the suppliers.

Internally, the Enterprise Resource Planning (ERP) system tracks all basic information to enable the management of the company's products such as handling orders and product stocks. Another system mentioned is the Product Information Management (PIM) system, which acts as an extended database to the ERP system including more technical data. This system is used for the case company's e-commerce. The respondent notes that information can only be transferred from the ERP system to the PIM system and not the other way around. This also implies that ERP information changes can only be made at the source, the ERP system. From a data quality perspective this is seen as a good latch. A respondent mentions that supporting technology through a process rather than having a process supported by technology is easier for the company to implement.

At OEM production sites, there are also different life cycle information tools to help customers track their products. A respondent mentions the usage of the company's serial numbers tagged onto the hoses but also other technologies that provide more information to the customer. The solutions discussed were QR codes and wrap around labels, the latter used as a standard in the marine sector. Respondents declare that OEM customers do not request the case company's tracking label as they often have their own solution to collect and trace data to which the case company complies.

The technology provided by the case company is a tracking label, a small plastic label attached onto the hose. As mentioned in section 3.2.2, this label tracks limited data using the serial number generated by the ERP system. Respondents report this solution as a good tool to track information and support customer service.

"The tracking label is a very practical tool for us to gather information and offer the customer more information" - Respondent

However, this label has its drawbacks and limitations. Even though the label itself is working well, the process to print this label is very troublesome and impractical, especially if implemented out in the field. The current solution requires internet connection, which is rarely accessible on customer sites, the printing sequence is stated by respondents to be "the wrong way around" and the printer often acts up. Furthermore, the label has its limitations in terms of physical aspects. For example, it cannot be implemented in too hot environments as the label burns or in very acid settings as the label deteriorates due to the chemicals. The labels are also very costly and not possible to reuse. Other technologies to update the tracking label are currently explored at the case company. Tools explored and mentioned by respondents are QR codes and RFID tags.

On site, some customers, often bigger clients, also have their own label solutions and systems for hoses. A respondent reports that it is often in the form of a cable tie and a small yellow label with an imprinted number. In other cases, the caps are marked through stamps or stainless ties are tightened onto hoses with the customer's own number. This is often the case in acidic facilities or in extreme

hot circumstances. An interviewee also notes that some of these customers had their own labelling system to track life cycle information before the case company's own label. This enables clients to keep track of their hoses and plan their own maintenance.

The service technicians working on sites are currently using very manual tools to collect and report information. Today, checklists and reports are filled out and made using paper and pen and later scanned or manually digitized.

"It is the case that you print out a service document and then travel to the customer and fill out that paper. And then maybe you write it down in a computer when you are back [at the store] or just scan the paper" - Respondent

To log quality deviations of products the ERP system is used today. There is no solution for structurally tracking product failures and other occurring problems connected to products. A respondent expresses the value of having data on past solved problems that you can look up.

"It would be worth it to have a case company Wikipedia where you can report such things." - Respondent

Overall, the goal at the case company is to digitize tasks more, including product life cycle information gathering. A new system with an app has been integrated to aid service technicians track life cycle information but is currently under development. Today, it is only used as an order system and no "valuable" information is tracked there according to a respondent. However, the vision is to create a thorough information gathering solution. The app gives the possibility to include a lot of information such as mapping a customer's entire machine center, machine, parts inside and other data. It is also supposed to include all documents and checklists that are mandatory for the service technician to fill out on assignments. The system also has an integrated AI function which could aid documentation, find information or even create summaries of previous work and assignments.

"You can fill out however much you like there!" - Respondent

Respondents see a lot of opportunities with this new system for multiple types of customers. Multiple interviewees have the idea of creating a platform that customers can access to see information or even add information themselves.

"We want to be able to provide customers a solution instead of just selling a product." - Respondent

Interestingly, respondents acknowledge that tools are necessary to simplify tasks and track information. Still, it doesn't matter how good these systems or tools are if the implementation and usage is not simple.

"It doesn't matter how good of a system you have or how good of a tool you have if it gets too complicated and troublesome." – Respondent

4.3 Opportunities and Challenges with Life Cycle Information

This section describes the opportunities, values and challenges that are seen with product life cycle information collection and traceability. It is discussed how life cycle information enables sustainable practices at the case company and what obstacles there are.

4.3.1 Opportunities and values with Life Cycle Information

All respondents see opportunities with collecting and tracing product life cycle information across a product's life span even though they are hesitant to how data gathering would work in practice. Interviewees mention a range of different kinds of values deriving from access to information.

A direct impact of information is the improvement of waste sorting. External respondents mention that the more product information, not to forget from the usage phase, the better opportunities are unlocked to handle waste in a beneficial way. Today, hoses are directly classified as hazardous waste, limiting these opportunities. Transport of hazardous waste is an additional service which also generates greenhouse gas emission and additional costs for the company. Through more information, products could be sorted better and unlock more recycling opportunities.

Related to the subject of waste, tracking information could enhance inventory management. Today, the case company has established routines that allow them to see how long a product has been stored through purchasing orders. However, they do not have a system currently that indicates or notifies when a hose is approaching the end of its shelf life. Employees sometimes inspects hoses visually to determine if they are still saleable when they are approaching the end of their shelf life, or if they should be sold at a reduced price or even be tossed. A respondent expressed that further information and data on the hoses in inventory can save time for employees, make inventory management more effective and reduce unnecessary waste by keeping track of products. For example, if there would be a system notifying of products approaching the end of their shelf life.

Respondents mention that the foremost reason to label hoses is traceability. Multiple aspects that can be improved through information, for customers, the case company and its employees, are highlighted as follows. The label is a very valuable tool when it comes to identifying problems, optimizing product lifetime and even avoiding operational downtime. Data helps hose maintenance in general, optimize planning of preventative maintenance and save time for customers and employees. A concrete example is that through the label the customer can call a store about its order and in turn, the product can be prepared in advance for pick-up. A respondent also gave the example that handling products with unpleasant content can be avoided by having a label, saving time and "dirty" work. A respondent also expresses that information can help optimize hose assembly lifetimes which in turn leads to benefits for the environment and saved costs for the customer. Out in the field, interviewees report that data can be a great support for service technicians and facilitate their work. Information, such as a problem description and material needed, can help prepare for planned assignments saving time and transport for both the client and the technician. Historic information can improve hose solutions by figuring out why the last one failed. The technicians themselves can also learn from past hose applications and improve their service accordingly.

Stronger customer bonds are another advantage that can derive from information. By developing the product into a service and asking more questions, information can enhance customer relationships and loyalty.

"People like to talk about their machines and such and if you can come and solve it and after that the hose doesn't break for 5 years. Then you have a customer for life and at the same time you have helped them." - Respondent

Respondents are also aware of upcoming laws and regulations to have more information on products sold. Collecting information as soon as possible gives the opportunity to work proactively and prepare for compliance and have an extra offering for customers before it becomes mandatory.

"Maybe we can create the rules?" – Respondent

4.3.2 Challenges when Gathering Life Cycle Information

One challenge mentioned by respondents is deciding how much and which kind of information should be gathered.

"How far should you track? Where do you draw the line?" - Respondent

Therefore, respondents mention that investigation of what kind of information is requested by customers is important. Otherwise, you risk gathering too much information which in the end is not necessary. Furthermore, respondents' reasons about the quality of the data gathered. Information collected from customers for example will never be a 100% accurate, which should be accounted for.

Respondents also discuss practicalities around life cycle information gathering. One example brought up is how willing stakeholders are to share information with the case company. The case company does not produce their own hoses or fittings and might face resistance from suppliers in sharing their exact material content. As the case company has a very large number of suppliers, communicating with them is challenging at times. While some respond quickly to requests, some do not, which can hinder life cycle information gathering.

The current life cycle information gathering technology, the company's tracking label, has some practical limitations that are discussed in 4.2.3. If such practical problems are not attended to, the case company risks missing out on life cycle information. Additionally, respondents mention the importance of employees' attitudes towards using life cycle information gathering technologies.

"In the best of worlds, you would want to have all the information available, but then I think it would be too much work. Then they might skip things on the checklist and just give the same answer for every question if there are too many questions on this checklist." - Respondent

The respondent further elaborates on the importance of considering the people working practically with the life cycle information gathering. The respondent argues that you should be careful with how you integrate such a process into the case company's existing operations, to make sure that it is feasible to implement practically.

4.4 Ways of Working with Life Cycle Information Gathering

This section describes the ways of working at the case company related to the gathering of life cycle information. It further discusses the challenges and opportunities with these ways of working. Additionally, this section includes needs expressed by respondents with regards to ways of working with information gathering.

4.4.1 Current Ways of Working with Life Cycle Information Gathering

Respondents mention that routines and ways of working with life cycle information gathering differ depending on location, store and employee. For example, respondents mention that not all stores across Sweden mention or try to sell the case company's tracking label to their customers, while some

always offer it. The decentralized organizational structure allows stores to decide for themselves, to a certain degree, what they offer their local customers and how they operate in their store. Stores have the possibility to bring in items from a local assortment at the case company and thereby adopt their offering. The decentralized organizational structure furthermore allows for a certain amount autonomy for the stores in their operational routines. Respondents mention that this makes it hard to know how one store operates in comparison to another one, as their routines differ.

4.4.2 Challenges with Current Ways of Working

The most common theme brought up by respondents when discussing ways of working with life cycle information gathering is the decentralized organizational structure. The fact that each country, market and store have a high degree of individual decision-making authority complicates the integration of common ways of working. Respondents mention that when routines and instructions are brought forward by the top management, it can be challenging to spread them to all operational countries and business units, as they have some authority in deciding how to integrate it into their specific context. This makes it hard to know in detail how each market operates, causing an immense number of exceptional cases, which can be costly in the long run.

Respondents mention that because the customer is what drives the case company forward, decisions are always based on what they request and need. This customer-centered mindset is highly valued by employees, and that sometimes requires going off routine and finding a unique solution. Therefore, respondents believe this must be kept in mind when integrating new ways of working. While there can be a common goal set by the top management, each market should get a say in how they interpret the goal and plan to integrate it into their specific operations. Respondents believe an overarching goal and vision, which has room for interpretation and adaptation, would be the most effective approach from the case company.

Moreover, it has been discussed by respondents that learning about hoses in general is difficult. Today, knowledge and practices are based on hands-on experience, learning-by-doing, and observations. Internal educations for employees about hoses exist, but it is reported that it is challenging to design these educations. Respondents report that employees can achieve a base-line knowledge through educations, but most of the knowledge is attained through learning-by-doing. Knowledge is said to come with experience. This causes employees to have a varying degree of knowledge regarding the products and operations of the case company depending on experience level.

When it comes to tracking life cycle information, respondents describe that the collection and storage of information acquired as somewhat unorganized. Once again, the routines differ between business units. Respondents mention that there is a need for established routines and systems for gathering and storing the information gathered.

4.4.3 Needs Related to Ways of Working with Life Cycle Information Gathering

The most clearly emphasized need by respondents related to ways of working with life cycle information gathering is that the processes need to be easy and integrated smoothly into current operations. The processes of tracking life cycle information should not feel like an extra burden for employees at the sales point in stores. In general, several respondents stress the fact that communication between decision makers and the rest of the company can be enhanced. They believe that to successfully implement new ways of working, the employees dealing with the practical implementation will benefit from receiving clear guidelines.

Respondents believe that it is important to communicate the purpose and benefits of tracking life cycle information to everyone involved in the process. Everyone from decision makers at the highest instance to service technicians installing products at the customer should feel clear about why they should spend time on tracking life cycle information, to successfully implement such an initiative. Therefore, when educating employees on new processes, ways of working or initiatives, respondents believe the case company should focus on communicating the value of working in this manner and the benefits it will lead to.

"You have to keep in mind how we want to launch this and how we talk about it and train people in it, make sure that there is a win for everyone, the customer and the whole company. It's about inspiring us to do something new instead of forcing someone to do something new." - Respondent

A respondent working with the supply chain also mentioned that ways of working around life cycle information gathering should be as standardized as possible to gain efficiency and structure. Although the respondent understands that it will not be possible to completely standardize ways of working, as the company is very large and operates in many different markets, they believe the case company could benefit from aiming towards this.

5 DEVELOPMENT OF THE LIFE CYCLE INFORMATION GATHERING PROCESS

This chapter presents the development of the thesis result – ***the Life Cycle Information (LCI) Gathering Process.*** *Different factors and aspects affecting the life cycle information gathering process in the various life cycle phases are also presented. The material presented here stems from the theoretical framework, interviews, study visits and discussions with industrial supervisors at the case company.*

In this chapter the problem is first defined more precisely to start the development of the concept. The next section shows the analysis of the problem, creating a foundation for the LCI Gathering Process. The concept starting point is then presented, followed by its concept development.

5.1 Problem Definition Used for Developing the Concept

Before developing a solution, it was necessary to understand and pinpoint the specific problem that had been identified through interview data. The interview analysis showed that collecting and tracking product life cycle information across products' entire life spans is not something that is performed all over the case company today. Nevertheless, certain product information is gathered in different life cycle phases using various systems that are not connected. This is making it hard for the information to flow as it lacks data and ties linking the existing data. The interview analysis made it apparent that the case company lacked structure when collecting information. In the literature study performed by the authors, there are limited product life cycle information gathering processes found.

Connecting back to the second research question in section 1.4, “What information gathering solutions are valuable to enable sustainable practices for an international hose assembling company?”, and the interview analysis, it was decided to develop a **Life Cycle Information (LCI) Gathering Process.**

Prior to the development of the LCI Gathering Process, factors affecting this process needed to be inspected. It was identified that the following four factors affected life cycle information gathering in each life cycle phase:

- **WHAT** type of information needs to be gathered? – Considering which information enables circular practices, what information is possible to track and what information should be prioritized.
- **WHEN** will information be gathered? – This needs to be considered both according to which life cycle phase but also within each phase.
- **HOW** will information be gathered? – Referring to what process, which technologies and what systems as well as other practical information.
- **WHO** will gather information? – Taking into account what actors need to or if they can perform this data collection.

This problem definition gave directions for what the concept should include and guided what research needed to be performed. Presented in the following section 5.2, it was established to carry out the following tasks:

- Inspecting the case company's product offering and how product types affect life cycle information gathering.
- Understand the different product journeys and their life cycle phases to identify how data collection is influenced by the life cycle loop.
- Identify different actors and their needs across the product life cycle.
- Understand how information gathering is performed at each stage today, including what systems and technologies are used.

5.2 Analyzing the Problem – Understanding the Case Company

A thorough investigation of the case company, its products and other factors affecting life cycle information gathering was made. The purpose was to understand the problem on a deeper level as well as identify opportunities and challenges when it comes to life cycle information collection and flow. This analysis is presented in the following sub-sections.

5.2.1 Product Mapping

As briefly stated in section 1.2, the case company offers a wide range of products, currently offering a product catalogue consisting of over 20.000 articles. The purpose of this section is to provide an overview of the case company's product offering and give a deeper description of the product segments this thesis focuses on. The product selection affects the development of the life cycle information gathering process, especially the questions of *what* and *how*.

The case company offers products and solutions within the categories of hoses, hose connectors, pneumatics and pipes (for both hydraulics and pneumatics applications). Through a subsidiary, various machines utilized for assembly of hose structures are offered, for example cutting of the hose in smaller pieces and crimp fitting. The last category, accessories, entails a variety of tools, oils, lubricants and sealings. These are commonly found in case company stores as an add-on product.

The largest product category for the case company is hoses and hose connectors, so called fittings and ferrules. The success behind the case company lies in their skilled assembly process of these components, creating a so-called hose assembly. Hoses need to withstand harsh environments at the point of installation, while at the same time transporting various kinds of media from one point to another in both high- and low-pressure situations. Therefore, both the quality of the components individually and the quality of the assembly of them play an important role in ensuring a tight and leak-proof connection.

Hoses can be further categorized depending on whether they operate in high-pressure or low-pressure applications. **High-pressure hoses** represent the case company's biggest source of income. These types of hoses are utilized for applications with normal to extremely high pressures, high impulses and high temperatures. As high-pressure hoses often operate in demanding environments, they are designed to be highly resistant towards wear, UV-exposure, high working pressures and pulsations. The high-pressure hoses are designed for application in hydraulics, but these types of hoses are also commonly found in high-pressure washing, chemicals, gas and spray paint applications. Therefore, there are several industries where they can be applied. Some examples are mines, the industrial

sector, cranes, lifts, agriculture, transport, offshore and forestry. See an example of a high-pressure hose application in an excavator in Figure 5.1.



Figure 5.1: Example high-pressure hose application in an excavator.

Low-pressure hoses, or industrial hoses as they are commonly called, is an umbrella term for describing the hoses that are used to transport a variety of media in low- or medium-pressure applications. Unlike high-pressure hoses, which most commonly contain hydraulic oil, there is not one media that is more common than another. Some examples of media are water, air, oil, gasoline, food, steam, chemicals and gas. Depending on the media, the inside material of the hose needs to be adapted to withstand it. For example, when utilizing low-pressure hoses in the food industry, no particles can be released from the hose into the media, adding strict requirements to the hose material. Examples of low-pressure hose applications can be seen in Figure 5.2.



Figure 5.2: Example of low-pressure hose application. Left: low-pressure hose in sewage. Right: low-pressure hose in distillery.

High- and low-pressure hoses and hose connectors are the products that this thesis focuses on. The life cycle information gathering process is developed considering both these types of hoses and their hose connectors. This decision was taken as this category represents the majority of the case company's annual sales. Investigating how to improve sustainability at the case company, and especially for these kinds of products, would therefore have the largest possible impact. Additionally, hose assemblies carry more complexity compared to other case company products, such as pipes.

Hose assemblies consist of multiple materials and are difficult to disassemble, making circular practices such as re-use and recycling difficult today. Such practices would be easier to implement for pipes for example, as they mainly consist of one material and can easily be disassembled and even washed.

5.2.2 Insights From Study Visit at Store and Mobile Hose Service

Two study visits were performed during the thesis project. Both were conducted at one of the case company's stores in the south of Stockholm. The first study visit was performed early on in the project and was of an exploratory nature to gain an understanding of working procedures at a store. The second study visit aimed to explore the specific ways of working and investigating how life cycle information gathering could be integrated into working procedures for both store employees and service technicians working with the mobile hose service unit. Both study visits provided valuable insights into the working processes and needs of employees. These insights were mainly utilized to answer the questions of *who, how* and *when* of the life cycle information gathering process.

Study Visit 1 - Store

During the first study visit, the authors observed a regular day at a case company store. They were guided by an employee with 12 years of experience at the case company, and 6 years of experience at this store. The authors learned that working at the store is just as much about customer service as it is about knowing the company's products well. When a customer enters the store, they often have a broken hose with them which needs replacing. The employee mentions that most of the time, hoses break due to unexpected external impact or wear and tear. The environment at the point of installation, for example temperature, bend radius and sun exposure affect the hose lifetime. The employee mentions that most hoses break before their theoretical lifetime is exceeded due to such reasons. Therefore, advising the customer of which type of hose is ideal in their specific installation environment is a large part of the job. This process involves talking to the customer and understanding their requirements before offering a suitable hose assembly.

Once the customer and employee has agreed on the product selection, the employee proceeds with the assembly process. The hose is cut to the desired length; hose connectors are then selected and assembled with the hose through a crimp fit.

The employee mentions that each encounter with a customer is different. Sometimes, the customer is in a great hurry and does not have time for talking or is interested in getting advice on their product selection. A common scenario described by the employee is that the customer works in a small business with a limited number of machines, if one of them cannot be utilized due to a hose breakage, the customers want to get their operations up and running as fast as possible. Then, it is most important to get the customer their replacement product as quickly as possible. This was noted by the authors, as adding time consuming steps in the employee's interaction with a customer might not be feasible in this kind of situation. Other times, the customer is less short of time and can spend more time at the store. The employee mentions that for these types of customers, they can devote more time to troubleshoot and investigate why a hose assembly has broken.

The ways of working and the time available for store employees vary greatly based on what the customer needs on that particular occasion. Therefore, the authors were able to conclude that any added steps into their ways of working need to be optimized to be as time efficient as possible.

Study Visit 2 - Mobile Hose Service

The second study visit was a ride-along with the case company's mobile hose service. The mobile hose service assist customers at their location with emergent hose breakages. The service technicians drive a van which is fully equipped with the most common hoses and hose connectors. See Figure 5.3.



Figure 5.3: Mobile hose service van at a construction site.

Similarly to what was observed during the first study visit at the store, service technicians at the mobile hose service met a wide range of different customers with different needs daily. The service technicians describe their work as changeable and that problem solving is the foremost important skill needed to do the job successfully.

During the study visit, the authors followed along a service technician and observed three different jobs. Pictures from the three different jobs can be seen in Figure 5.4.

1. Infrastructure construction site

During the first job, a customer called the mobile hose service as their machine was leaking hydraulic oil from a hose. It was quite urgent to attend to this, as hydraulic oil is very environmentally harmful if it is absorbed into the ground. The customer was a hired contractor working on an infrastructure project in the Stockholm area. The customer had in turn rented the machine from another company. Therefore, they were keen to have the hose breakage repaired professionally by a service technician to minimize any liability issues.

2. Construction site

The second job was for a medium-sized construction company working on preparing the ground of a land area for a building. Once again, hydraulic oil was leaking from a hose on an excavator onto the ground. This customer owned the excavator themselves. The customer was keen to have this issue fixed as quickly as possible, as the broken excavator stopped them from continuing their work at the site.

3. Crane truck

During the third visit, a hose which was leaking hydraulic oil was exchanged on a crane truck for a medium-sized waste management company. This was a standard job for the service technician, he had exchanged the same kind of hose for this customer one week prior. It was less important for this customer to repair the crane truck quickly if compared to the previous customers, as the customer had multiple crane trucks available and could continue their work in parallel.

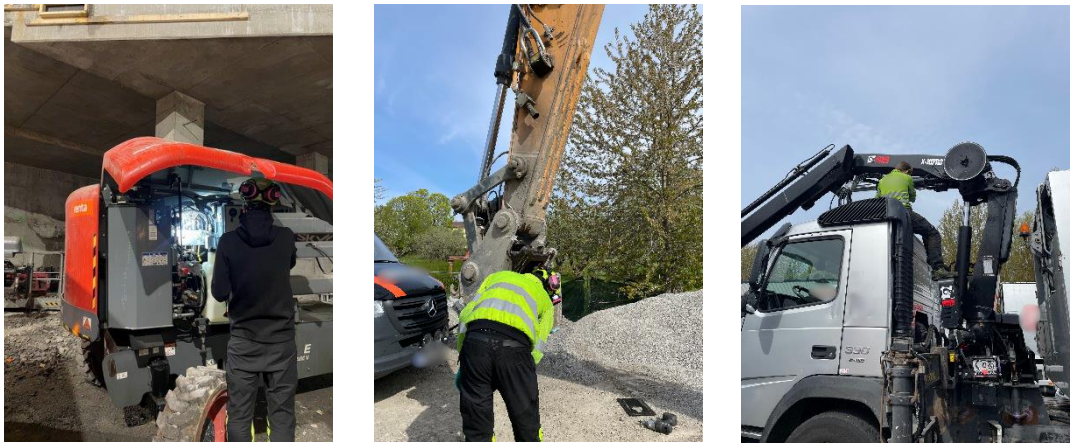


Figure 5.4: Pictures from the three jobs observed during study visit 2. From left to right: "Infrastructure construction site", "Construction site" and "Crane truck".

The main insights the authors made from observing these three different jobs were that a service technician, likewise the store employee, must adapt their ways of working depending on what the customer needs at that moment. While some customers had time to talk and explain the problem in detail, others just wanted the problem to be dealt with as quickly as possible to continue with their tasks. Therefore, the time available for service technicians to perform extra tasks, such as life cycle information gathering, varies greatly depending on the type of job. This provided insights into the questions of *how* and *when* in the life cycle information gathering process.

The physical requirement of the service technician's variates a lot depending on the job, as seen in Figure 5.4. They variate between climbing crane trucks to working 90 meters underground in a tunnel. The service technicians mention that the current tracking label system utilized by the case company is not optimal to use in all environments. For example, the service technicians mention that internet connection is required to be able to print the label before assembly, which is not always available if the mobile hose service must travel to an off-grid location. This also provided insights into the question of *how* in the life cycle information gathering process.

5.2.3 Competitor Analysis

To understand the competitive landscape for the case company, as well as investigating if life cycle information gathering is offered by them, a competitor analysis was made. The authors solely performed desktop research on the websites of multiple competitors with similar product offering as the case company. The researched competitors also offer product tracking solutions. Two competitors were chosen for deeper analysis. The competitor analysis provided the authors with insights into *how* life cycle information gathering can be done.

The analysis showed that both competitors offer tracking technologies and life cycle information gathering. Tracking of the products is conducted using various types of labels mounted on the hose at assembly. Information on the products can be found in a system, available both for employees at the competitor and their customers.

The main value of product tracking marketed by the competitors is the ability to optimize maintenance and reduce the number of unexpected breakages. For example, when a customer has tracked events connected to their hose assemblies for a longer period of time, the competitor can create a predictive maintenance plan based on the tracked maintenance history. This not only reduces the amount of emergency repairs but also reduces environmental damage due to emergent hydraulic oil spillage. Additionally, the information gathered on products allows the competitors to create plans

for recycling and reuse of their products. The more they know about the hoses, the better opportunities there are for investigating what the best option is for the hose at its end of life.

One of the competitors offers a subscription-based payment model for tracking products. Depending on subscription level, the customer receives different levels of service from the competitor. For example, on the basic level the customer can access the platform and view all their hose assemblies, but they perform inspections and maintenance themselves and record it in the system. If the customer pays for the pro-level, technicians from the competitor perform all inspections and maintenance, records it in the system and also provide a thorough risk assessment and maintenance plan.

5.2.4 Product Journey Map and Scenarios

To understand the life cycle of the case company hoses, their life spans were mapped out by creating a product journey map. This activity was inspired by the classic customer journey mapping exercise, made to explore and outline the stages a customer goes through using a product or service (van Boeijen et al., 2013). In this case, the journey of the product in different types of usage scenarios was analyzed to determine the product and material flows. The journey includes all stages of the products life cycle loop that were identified by the authors. The aim was to pinpoint where information gathering could occur, stakeholder interaction points where they could contribute with information as well as obstacles for information gathering in each stage. This activity links back to answering, mainly, the questions *when* and *who* from the problem description in section 5.1.

Three types of product flow scenarios were identified: *Private customers or small B2B purchases, Large customers* and *Large installations*. The first scenario called *Private customers or small B2B purchases*, describes the journey of a hose assembly purchased as a spare part for existing products. As the name indicates, the users in scenario 1 are private customers or clients making smaller business to business purchases. It is important to underline that the product is used as a part in an already existing product, such as a tractor or machine. In contrast, the second scenario depicts a hose assembly used as a component in the production of a new product. This is often the case for the company's OEM* productions. The users are in this case large industry customers, for example from the automotive and machinery industry. The final product is sold to another customer or end-user making that product loop complex. The third and final scenario uses the hose assemblies as a spare part for existing installations. In scenario 3, the users are large industrial installations and production centers such as oil rigs and large industrial manufacturing factories. The product is used at the site of the customer. An overview of the 3 scenarios is given in Table 5.1.

* An OEM produces products or components that are sold to other companies to be integrated in their own products.

Table 5.1: Overview of the three product journey scenarios.

Scenario	1	2	3
Short description	Hose assembly purchased as spare-part for existing products	Hose assembly used as component in the production of a new product. Often OEM production	Hose assembly used as spare-part for existing installations
User	Private customer or smaller business client purchases	Large industry customers (e.g. from the automotive industry, machinery industry, etc)	Large industrial installations and production centers (e.g. oil rigs, manufacturing center, etc.)
Usage Type	Product is used in existing product (e.g. tractor, machine, etc.)	The new product is sold to another customer or end-user	Product is used at the site of the customer

The different life cycle phases were mapped out for each scenario type in a product journey map which can be seen in Figure 5.5.

In *scenario 1*, the components from the hose assembly travel from the manufacturer trough the case company headquarters to a store or directly to a store. At the store, the hose structure is assembled for the customer at purchase. After purchase, the hose assembly is used until it needs to be replaced, either due to failure or for maintenance reasons. The customer can either contact the case company after market service to replace the hose assembly or return to the store directly. In the case of the after-market service contact, a new hose assembly is installed on site and the used hose is returned by the service technicians to the closest store recycling station. If the customer goes directly to the store, the used hose assembly is oftentimes brought with to aid the product replacement and handled by the store. If not brought with, the authors have been instructed that the customer gets rid of the used hose independently. In both cases, the customer buys a new hose assembly and uses it until the cycle repeats itself. The used hose at the store is sent to the case company's waste management partner. From there the authors expect the hose assembly to be recycled so that the material can enter a new product loop or that the energy created is used for other purposes. What happens after the hose assembly is sent to the waste management partner is unknown to the authors despite investigation.

The unknown events in the life cycle phases increase in *scenario 2*. Here the products are sent from the supplier to the case company production site directly or through the case company warehouse. The products, both in assembled form and separate parts, are sent to the customer often in large volumes. The large customer implements the case company products into their own final products, for example a car or an excavator, which is further sold to another user. If the hose within the final product needs to be replaced, the large customer sometimes has its own after-market service, or the customer can visit a case company store. A new product is provided in both cases. If replaced at a case company store the waste enters the final stages of the loop in scenario 1. What happens to the hose if replaced through the large customer after-market service is unknown to the authors. Potentially, these actors have their own waste management system. Note that the final product, which includes hose assemblies, provided by the large industry can be further sold from one end-customer to another where the product journey becomes even more complex and unknown.

Scenario 3 shows the product journey when the hose assembly is used in a large installation. The product travels from the supplier to a case company production center, where it is assembled and then sent to the customer at its large installation. If the hose assembly fails or needs to be replaced,

the case company assists the client via their after-market service or sends a new order from the production center. An interviewee mentions that the large installation clients potentially have their own waste management partners as their waste is not received unless the after-market service is involved. In the latter case the waste is returned to a store.

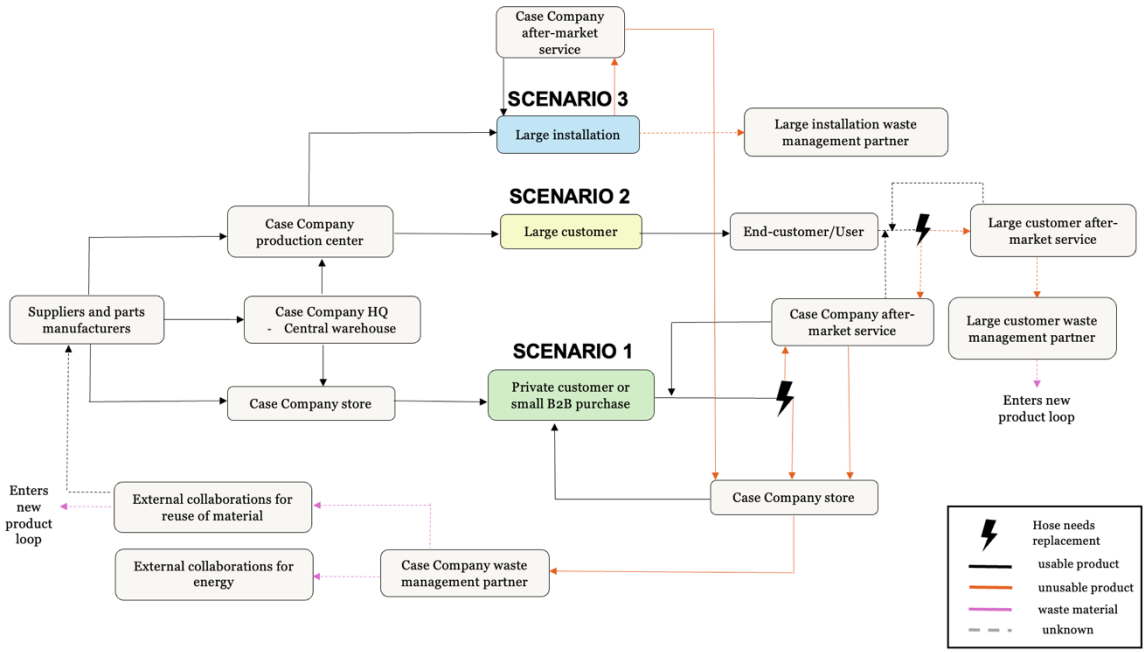


Figure 5.5: Product Journey Map illustrating the three scenarios and their product flows.

5.3 Concept Starting Point

This section provides the foundation and starting point for the LCI Gathering Process, which could also be referred to as the frame of reference for the concept development. The first and foremost base presented is the chosen product journey scenario, see section 5.2.4, and its life cycle phases. Following, stakeholders and their needs across the life cycle phases are described, along with limitations for the concept development.

5.3.1 Chosen Product Journey Scenario and Life Cycle Phases

Scenario 1, *Private customers or small B2B purchases*, was chosen as the focus product flow for this thesis. This choice was made as nearly the entire life span of products could be identified and knowledge about what happens in each stage was most accessible, but also because the case company had the highest control and opportunities to collect product life cycle information in-house in this case. Furthermore, the authors found that this scenario to have the least obstacles and be the most feasible due to the thesis project limited time frame compared to the other scenarios which required extensive collaboration and reliance on external stakeholders.

Figure 5.6 shows the product loop and its defined life cycle stages. To show the journey of the product, each phase is presented with a corresponding description in Table 5.2 .

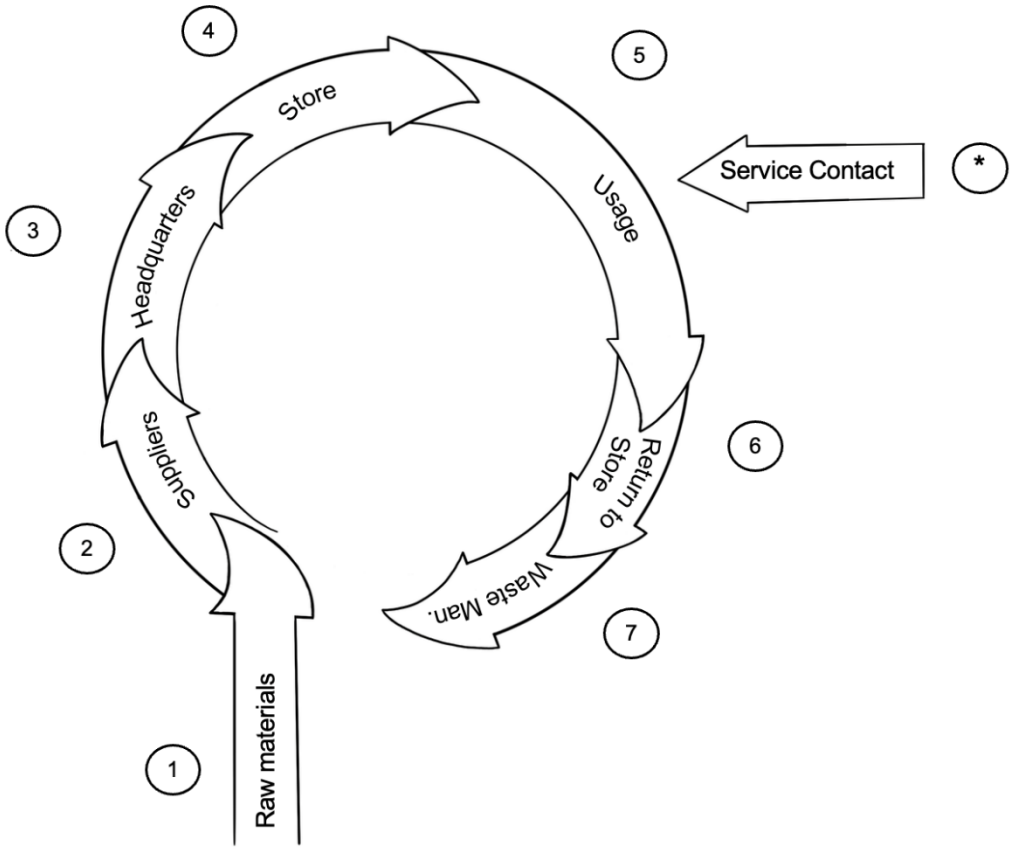


Figure 5.6: The product life cycle stages (from scenario 1).

Table 5.2: Overview of the three product journey scenarios.

Product Life Cycle Stages	Description
1. Raw Material Extraction	Raw material is extracted and sent to supplier for product manufacturing.
2. Manufacturing at Supplier	Supplier manufactures products according to orders and demands from the case company. Products are sent to the case company headquarters warehouse or directly to the case company stores.
3. Case company Headquarters (HQ) and Warehouse	The headquarters, incorporating a warehouse, are in contact with the suppliers, OEM centers and stores. This place receives parts and limited product data from the suppliers. It sets guidelines for the case company's operations and transfers knowledge though the organization.
4. Case company Store	The store either receives products directly from suppliers or from the case company headquarters. This is the customer contact point, in the form of purchase, service and/or advice. The case company is decentralized, and stores have a certain amount of freedom in deciding on their ways of working. Stores oftentimes communicate with other stores and are the base for the mobile hose service. The mobile hose service refills products and materials needed for jobs as well as return waste and unused material from finished jobs.
5. Usage	The usage phase describes the period of time when a product is used by a customer. The products are installed and used in a range of different applications.
*Service Contact	Service contact occurs when a customer needs servicing or repair of products. The product is replaced by a new solution and the used one is taken care of by the service technicians and returned to in-house waste management centers at stores. The mobile hose service operates 24/7 and is often used for emergencies but also for maintenance jobs.
6. Return to Store	In the case of product replacement, the customer returns to a store. Oftentimes the customer brings the used hose to the store to aid the replacement. In this way, the used hose can be sorted at the in-house store waste facility.
7. Waste Management	The waste from the store, including used products, is picked up by the waste management partner of the case company. Here, the product is turned into energy or recycled.

5.3.2 Stakeholder Needs Across the Life Cycle Phases in Scenario 1

To develop a feasible life cycle information gathering process, stakeholder characteristics, which affect stakeholder needs, had to be determined across the life cycle phases. The identified needs were important to answer the third and the fourth factor, *how* and *who*, affecting information collection. Through the interviews and study visits, characteristics for customers, employees and service technicians were identified.

Customer types and their characteristics

Customers affect the life cycle information gathering process as they are the main actors providing the information prior, after or in the usage phase. Customer attitudes and willingness to provide information vary as well as their knowledge level when it comes to information. Accordingly, three main customer types were identified in Scenario 1: "Private customer", "Small business" and "Customer has rented a machine". These types of customers vary according to the amount of machinery in possession and machine ownership status. When it comes to characteristics, two out of three customer types care about cost. This is often related to a good price to quality ratio. Furthermore, the types of customers perceive the value of qualitative service and the amount of time needed to solve the problem differently. The different customer types and their characteristics are described in Table 5.3.

Table 5.3: The three different types of customers and their characteristics.

Type of customer	Private customer	Small Business	Customer has rented a machine
Description	Hobby mechanic who repairs his own motorcycles, machine or other product needing a hose assembly.	Small business with 3-10 machines.	Larger contractor who rents machines from a 3rd party.
Characteristics (factors are ranked from the top, the top being the most important)	• Cost Sensitive– reasonable price quality ratio • Values Good Service – advice of product, hose assembly, selection	• Time Efficiency - Short time between problem discovery and measure • Cost Sensitive– reasonable price quality ratio • Values Good Service – advice of product, hose assembly, selection	• Requires a "Correct Solution" - Job done " according to the book" • Time Efficiency - Short time between problem discovery and measure

Store employees and their needs

Employees at the store play a key role in collecting life cycle information. As mentioned in section 4.4.2, the employees at the case company rely mainly on tacit knowledge acquired through learning-by-doing and experience. It was concluded from the study visits and interviews that employees have different levels of knowledge and ways of working depending on their work experience. It was identified that such factors affect the attitude and willingness to collect life cycle information gathering in different ways. It was deduced that an employee with less experience follows routines to a greater extent than an employee with more experience. An employee with greater experience is more likely to deviate from routines as they have found their own ways of working through more experience. Following these insights, employees were classified in 3 categories according to their experience. The classification is an analysis which stems from interview data and observations at study visits. The classification and the corresponding characteristics can be seen in Table 5.4.

Table 5.4: Employee characteristics according to experience.

6 months to a year	2 to 5 years	5+ years
Instructions and guidelines are important Focus on doing the tasks "right" rather than fast Often ask for help and advice from more experienced co-workers	More self-sufficient after 2 years of experience Efficiency is appreciated "How can I work faster and more efficiently?	Work fast paced and streamlined Optimized their way of working More knowledge: Can take their own initiatives & more advanced decisions Think out of the box: "How can I improve …?"

Service technicians and their characteristics

At service contact, service technicians are the actors performing the job on site. This means that, in that event, service technicians are the only reliable actors that can gather information. From interviews and study visits, three main components when performing a servicing job were concluded. The first one being that the service technicians need to be ready to solve all kinds of problems. "It is more than just changing a hose." Secondly, service technicians need to be able to adapt to uncertainties and be agile to save time and provide the best possible service. Finally, it was observed that communication between service technicians was essential and needed to be accessible at all possible times. Today, this communication is often performed in the form of phone calls.

Apart from these common needs which apply to all jobs, the type of assignment lays the foundation for more specific needs. Service jobs were categorized into three categories – *Emergency*, *Maintenance* and *Planned Projects* - and differentiated according to number of tasks, time and information characteristics. Table 5.5 shows the three types of service jobs and their characteristics according to the factors mentioned previously.

Table 5.5: Types of service jobs and their characteristics.

Type of Job	Emergency	Maintenance	Planned Projects
Number of tasks	Emergency tasks vary depending on the situation. It can be a smaller task or a bigger task.	Smaller tasks are often fixed in one session	Multiple tasks can be given to fix over multiple days.
Time characteristics	Time is of essence. Jobs are urgent to fix.	More time is available - Less urgent.	Time frame is set by customer.
Information characteristics	As much information as possible about the job in advance facilitates task but not often given. Information about stock in stores is also beneficial.	Information is oftentimes provided by customer.	Higher safety instructions are given, and more equipment is often required.

5.3.3 Limitations for Concept Development

The authors of the thesis project decided to focus on the following life cycle stages in the chosen scenario: *Case company headquarters (HQ), Case company Store, Usage, Service Contact* and *Return to Store*. This limitation was made due to the thesis project limited time frame but also because of the dependence on external stakeholder information in the remaining stages – *Raw material extraction, Manufacturing at supplier* and *Waste Management*. An additional reason for this choice is that these focus life cycle stages are the phases in which the case company has the least obstacles to a life cycle information gathering process implementation. They are in control and can decide to apply such a practice within the company. The focus stages are highlighted in the life cycle chart in Figure 5.7.

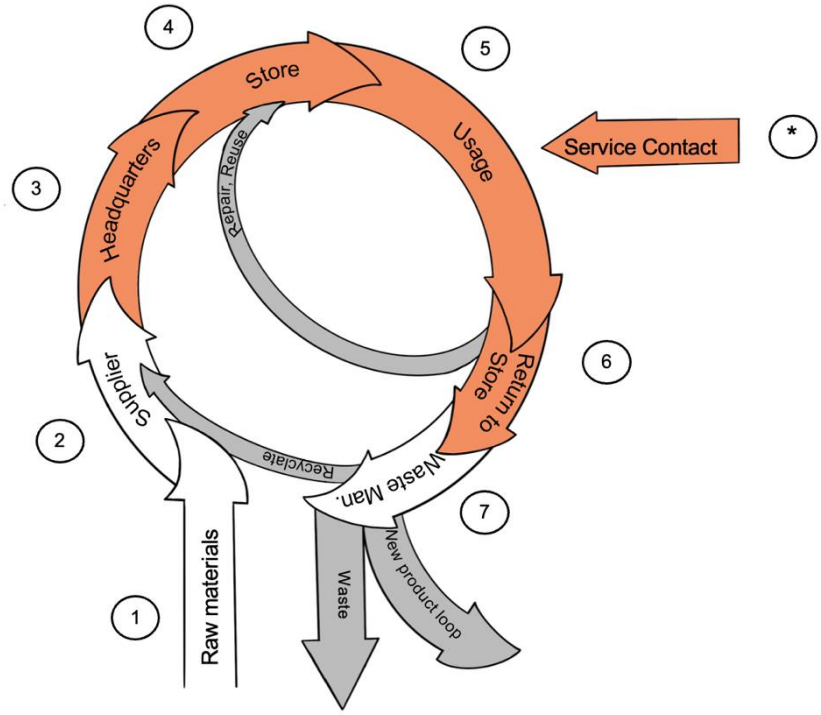


Figure 5.7: The focus life cycle stages in the product loop.

The thesis is also limited to developing a process and does not investigate deeper into technologies and systems. This decision was made in conversation with the case company, expressing the need for a strategy and process rather than implementation of new technologies and systems. It was important to them that a process was developed prior the technology instead of the other way around. The latter case had been observed before, for example with the tracking label printer, not giving ideal results. See more details in chapter 4. Furthermore, systems and technologies might be established for life cycle information gathering in accordance with the DPP regulation (see chapter 3.3 and 3.6), meaning that changes in such might occur soon anyhow. Moreover, creating an entirely new system comprising all exiting systems or adding a new system to the currently unliked and diverse systems did not align with the purpose of the thesis project. Changing such a big system would require Information Technology (IT) knowledge and major shifts in the case company.

5.4 Ideation and Concept Development

After a foundation for the life cycle information gathering process had been established, the concept was developed. This included an ideation phase with a workshop made together with multiple employees at the case company and a brainstorming activity with the authors alone. The aim was to explore the factors of *what, when, who* and *how* defined in the problem definition, see section 5.1, but also determine what factors were to be implemented in the final result. Selection criteria and factors are presented in this section along with the activities mentioned above.

5.4.1 Ideation workshop

Halfway through the thesis project, in April 2025, a collaborative workshop together with multiple employees at the case company, including industrial supervisors, was conducted. Detailed workshop material can be found in Appendix B. The purpose of this workshop was to generate ideas related to

the questions of *what, when, who* and *how,* which were analyzed and used to create criteria for the result.

The workshop was divided into two parts. The first part aimed to ideate around life cycle information gathering on the entire life cycle of a hose, while the second focused solely on the usage phase. The workshop participants first brainstormed individually, writing down their ideas on post it notes before discussing their ideas in small groups of three people. At the end of the workshop, the groups wrote a pitch for their ideas and pitched them to the other groups. In total, there were eight workshop participants divided into three groups, creating two groups of two participants and one group of three. The pitch sheet can be seen in Figure 5.8.

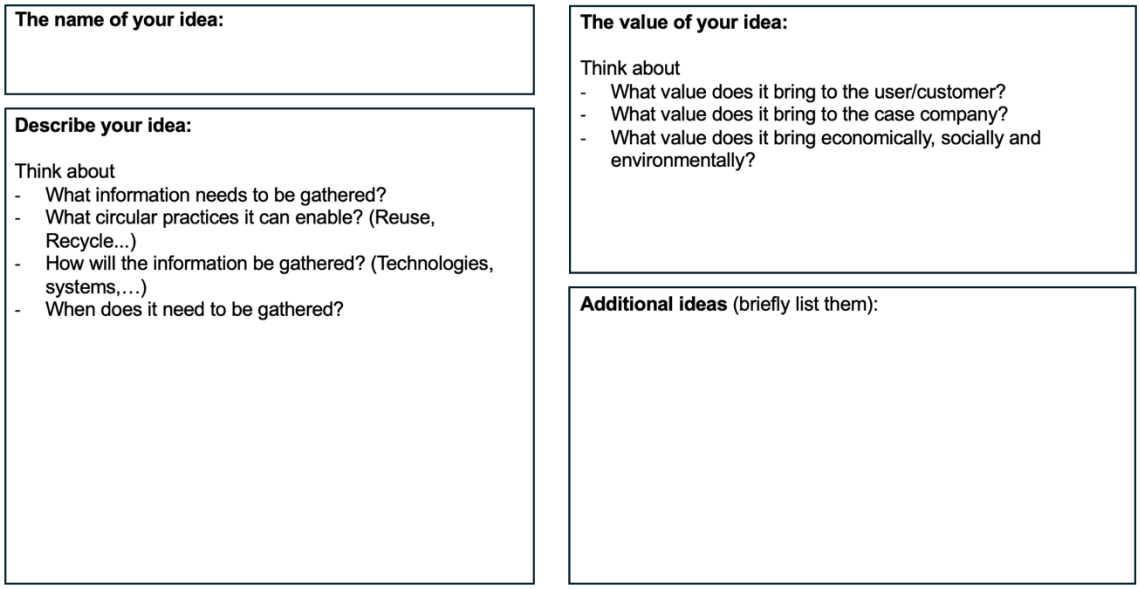


Figure 5.8: Pitch sheet provided during the workshop.

The results from the workshop are presented in Tables 5.6 -5.8 below, representing the three workshop groups.

Table 5.6: Results from workshop group 1.

Question	Idea
What information needs to be gathered?	All supplier data (e.g. origin), internal sustainability data, health effects
What circular practices can it enable?	Recycling information
How will the information be gathered?	Gather data from different sources and create a common system where it can be accessed by customers and case company employees. Data should be gathered using QR-codes and accessed in a app.
When does it need to be gathered?	When gathering data from customers: before the purchase. When gathering data from suppliers: as early as possible. It should be a requirement for suppliers to provide data.
Value of the idea	Fulfilling ISO-standard requirements.

Table 5.7: Results from workshop group 2.

Question	Idea
What information needs to be gathered?	Media, application, working pressure of hose, operating temperature, operating time, details of environment at installation point.
What circular practices can it enable?	*None stated.*
How will the information be gathered?	Through a digital platform. The customer can register which parameters are important for them, for example price, time, sustainability etc.
When does it need to be gathered?	Upon purchase.
Value of the idea	Safety and operational security for customers. Time.

Table 5.8: Results from workshop group 3.

Question	Idea
What information needs to be gathered?	Media, working pressure and temperature.
What circular practices can it enable?	*None stated.*
How will the information be gathered?	The hose has a QR-code that can be scanned, this will take you to a digital platform where information can be viewed.
When does it need to be gathered?	Upon purchase.
Value of the idea	Feedback loop to the case company. It is easier to re-purchase the same product. Service intervals are easier to establish.

Following the workshop, the results from the workshop were synthesized and analyzed by the authors. The results from the workshop were beneficial in the final brainstorming and selection process, described in section 5.4.2.

5.4.2 Brainstorming and Selection

After gathering and exploring options to answer the questions of *what, how, when* and *who* in each life cycle phase, a final brainstorming session was performed by the authors before selection. In this brainstorming activity, the authors mainly ideated on the question *what*, as it was the least clear and explored questions.

To decide on the final content of the result, the authors analyzed the material and selected answers to each category according to the following requirements:

- The process should fit as seamlessly as possible into the case company's current operations

Both case company employees and their customers experience situations where they are short on time, for example in an emergent breakage of a hose. Therefore, adding time-consuming steps into their working processes were to be avoided.

- The act of gathering life cycle information should be easy and simple

All stakeholders involved in the life cycle information should find it as simple as possible to take part in the process.

- The type of information gathered should be easily accessible for both customer and employee

For example, gathering data on the number of operating hours for a hose was discarded due to complexity. It would require the customer to provide data on the type of machine and operating hours of it, which is not easily accessible in all cases.

- Implementation is feasible

The process suggested should be able to be implemented into the current operations of the case company.

6 FINAL CONCEPT AND RECOMMENDATIONS

*In this chapter the final concept – a **Life Cycle Information (LCI) Gathering Process** – is presented. In addition to the concept overview, values and benefits with life cycle information gathering are discussed. The chapter also includes overall recommendations for a CE transition in accordance with LCI collection. Implementation challenges and possible obstacles are also determined. This chapter links back to the third and final research question "What opportunities and risks would come with such a solution?". The concept answers the second research question "What information gathering solutions are valuable to enable sustainable practices for an international hose assembling company?". This chapter is a result of all data collected in this thesis and analysis made.*

6.1 Final Concept Description

A result of the master thesis is the **Life Cycle Information (LCI) Gathering Process** to support the information collection at the case company. The process is intended to help the case company collect product life cycle information by providing guidelines on *what* information should be gathered, *when* the information should be gathered, *how* information will be gathered and *who* will gather the information. Due to limitations, see section 5.3.3, the process describes life cycle information gathering in detail in 4 stages – *Case Company Headquarters and Warehouse, Case Company Store, Usage & Service Contact and Return to Store* – and gives general recommendations for the remaining life cycle phases. The product and material flow for the LCI Gathering Process can be seen in Figure 6.1 and an overview of the process answering the question *what*, *when*, *how* and *who*, can be seen in Table 6.1. Each stage and its sub-process of gathering information is described in detail the following sub-sections 6.1.1-6.1.4.

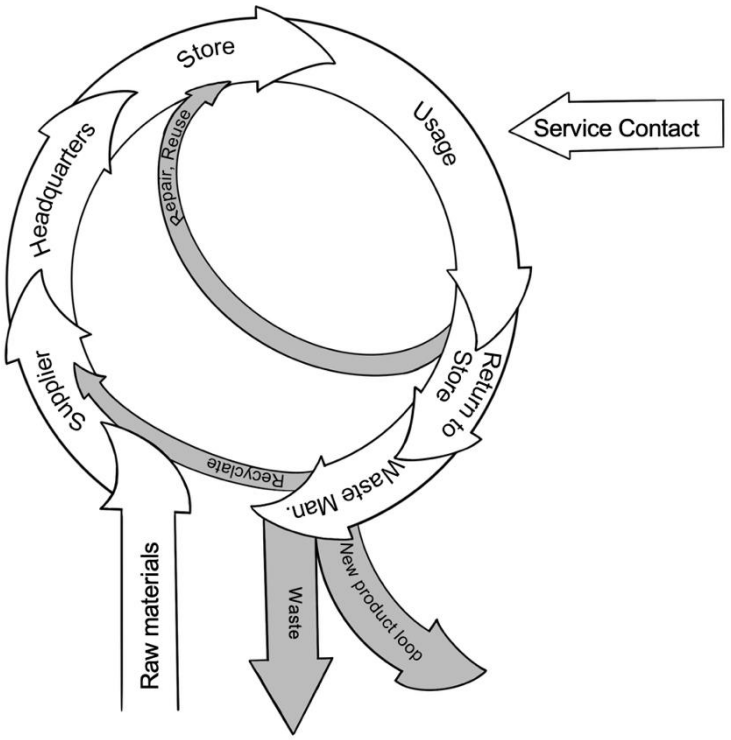


Figure 6.1: The LCI Gathering Process product life loop for a hose assembly.

Table 6.1: The LCI Gathering Process in each product life cycle phase.

Product Life Cycle Phase	***What*** information should be gathered?	***When*** *should* information be gathered?	***How*** should the information be gathered?	***Who*** should gather the information?
Case Company Headquarters (HQ)	Detailed material composition Manufacturing process Production date and time Guidelines for usage (e.g. product manual and safety data sheet) Regulatory compliance documents (e.g. material) Environmental footprint (from manufacturing and material extraction) *Information ideally attained from case company suppliers.	When products are purchased from supplier.	In a digital in-house platform.	Employees at the case company headquarters.
Case Company Store	*Information about a new hose:* Hose structure Product batch Product assembler (which employee) Predicted media that will be used	During purchase but before payment.	Information gathered verbally and documented in digital in-house platform. Ideally connected to the tracking label's digital platform.	Store employees.

	Description of environment at installation Reason for hose exchange Inspection and maintenance history Any other mentioned information			
Usage Phase	Currently no information will be gathered.	-	-	-
Service Contact	*Information about a new hose:* Hose structure Product batch Product assembler (which employee) Predicted media that will be used Description of environment at installation Reason for hose exchange Inspection and maintenance history Any other mentioned information *Same information as gathered at the store.	During or after service visit depending on type of job.	Information is attained verbally or through observation and documented in a digital platform.	Service technicians.

	Information from the used returned hose: Used media Description of installation environment Reason for hose exchange Hose lifetime *Same information as for a used hose returned to a store.			
Return To Store	*Information from the used returned hose:* Used media Description of installation environment Reason for hose exchange Hose lifetime *Checkpoint from previously gathered information at purchase.	Upon product return.	Information attained verbally and documented in in-house digital platform.	Store employee

6.1.1 Life Cycle Information Gathering at Case Company Headquarters and Warehouse

As mentioned in Table 5.2 the Case Company Headquarter is the heart of the Company and acts as a warehouse. It decides what products are in the group* assortment and is in contact with multiple stakeholders and company units such as the suppliers, OEM centers and stores. Today, the headquarters receive limited product data from suppliers and use multiple systems, the ERP and PIM system internally and external tools, to record product information in-house and display information to customers.

* The company has a general "group assortment" which is the same throughout company units. Additionally, each location has their "local assortment" varying according to the place's wants and demand.

Contrarily to today's ways of working, the process suggests establishing tracking initial product information when products arrive from suppliers. Types of information suggested to be tracked are presented in Table 6.1. Ideally information should be provided by suppliers. However, if that is not possible the case company is strongly suggested to investigate such and collect information independently. Note that certain information may be reoccurring if the same product is purchased multiple times. The table also mentions the idea of having an in-house digital platform. Ideally, one system should be used to gather life cycle information. Today, the existing systems are suggested to be used, and preferably connected, to establish the habit of gathering information. However, investigating and developing a system solution was not part of the project scope. The authors also recommend appointing a specific managerial role for overseeing this process. Multiple employees can be responsible for gathering information.

6.1.2 Life Cycle information Gathering at Case Company Store

At a case company store, the new hose is sold to a customer. A tracking label should be put on all sold hoses and the store employees ask questions to gather and document information. To assist the types of information stated to be collected in Table 6.1, the following questions have been formulated:

- What media do you expect will flow through this hose?
- What environment will the hose be installed in? Hot? Cold? Other important external factors?
- What is the reason for buying this new hose? Replacement? Why did the previous hose fail?
- Has any maintenance or inspection been carried out by the mobile hose service?

An additional free text field should be added to write down other information deemed interesting by store employees. The employee documents the data before payment. The system should be built so that payment cannot be made without attempting to track information. If the customer cannot answer the questions, fields can be marked with "unknown". The purpose is to implement a habit for employees to track information but also expose customers to life cycle information gathering. Long term, the customer is more aware of upcoming questions and can contribute with more and more information.

6.1.3 Life Cycle information Gathering during the Usage Phase and at Service Contact

In the usage phase, it is not suggested to collect information because of the multiple challenges that follow. An obstacle is, for example, the lack of systems accessible for customers to track information. In the future, customers can possibly participate in information gathering by documenting continuously during usage in an accessible and user adapted platform.

Table 6.1 describes what, when and how information will be gathered by service technicians at a service contact point. The process is relatively agile due to the variating natures of service jobs.

6.1.4 Life Cycle Information Gathering as the Product Returns to Store

When a hose needs to be replaced, it is returned to the store when a new hose is purchased. At this stage in the life cycle, a checkpoint is established for the used hose to check if information provided at purchase has changed. The purpose of this is to enhance data reliability. The information might also be entirely new if the used hose was purchased before the information gathering process was implemented. If anything has changed from the earlier prediction, it should be updated in the system.

The store employees should also ask about hose lifetime in this step. The new hose that is sold enters the process from the case company store phase.

6.2 Values and Benefits with Life Cycle Information Gathering

The values and benefits of gathering life cycle information were inspected with regards to impact and time frame. A simple categorization was made between the values acquired from life cycle information with minimal effort to benefits that arise with more time and investment of resources. Most immediately, it was identified through interviews that waste management could be improved significantly with greater usage information. Environmentally and economically, hazardous waste pick-ups, which are made in addition to "regular" waste pick-ups adding up to a significantly greater cost, could be minimized and recycling enhanced. With further investigation and development, the case company could expand their value offering through greater servitization, such as maintenance or subscription models. Sustainability work also includes benefits for the company competitively, as customers are kept and new customers are attracted. This was mentioned on multiple occasions during the interviews, see section 4.1.4. Being transparent and communicating sustainability actions also fosters stronger customer bonds and retention as well as creating a marketing opportunity.

Over time, with more accumulated data, options for better recycling can be explored. Data collected can provide directions for material amounts which motivate investments to explore such. Reuse possibilities can also be inspected. Additionally, through information exposure, users accumulate knowledge helping them make better usage choices. Which in turn benefits the customer economically and decreases the amounts of products wasted due to wrongful handling.

With increased data, product lifetime estimates can be made more accurately. Accumulated hose failure data can be used to enable such. Through extensive usage information, design improvements can also be identified to optimize products. This benefits the company as they can provide the best solution for their customers, simultaneously the user is satisfied with better product performance and lower costs over time.

One could argue that the greatest economic value is that information gathering prepares the case company for future regulations, mainly the DPP, and in turn helps avoid extensive fees. With extensive data the company can also improve their stock management, analyzing what and how much is sold to avoid waste. Note that the demand for products in the hose industry varies and was said to be unpredictable during the study visit. It was mentioned that seasonal predictions can be made and would be improved through information.

Finally, the most long-term value is increased collaboration possibilities fostered by information sharing. Overall, increased possibilities for circular strategies are unlocked such as efficient transport and the creation of a take-back system to make the product and material flow even better. Figure 6.2 gives an overview of the classification of values.

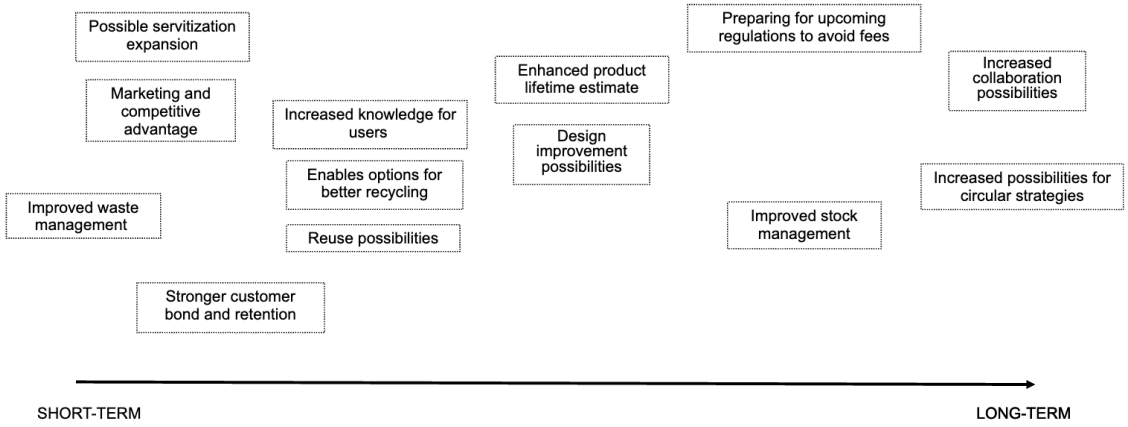


Figure 6.2: User benefits as well as business and environmental values with product life cycle information.

6.3 Overall Recommendations

Along with the LCI Gathering Process a set of recommendations derived to help the case company in its transition towards circularity. The recommendations are presented below:

Make sustainability goals and definitions more accessible for all employees at the case company

From the interview analysis the authors identified that not all employees at the case company are aware of the established sustainability goals and definitions stated by the case company. Therefore, it is recommended that the case company puts more effort into making sustainability goals and definitions more accessible for all employees in order to spread awareness throughout the company.

Spread established vision and mission across all company units

On the same topic, specific goals should be set for each case company unit. Clear action plans can help understand and showcase how to work with sustainability, which through this thesis has been proven not to be crystal clear. The aim is to spread a common mindset across the company to ensure an equal sustainability knowledge level, which assists the shift towards circularity. This shift requires to implement the mindset into the core values of the company and get everyone on board. Meaning that each employee in the organisation shall be aware of the set sustainability vision and know how to apply it practically in their unit.

Implement Life Cycle Information Gathering using the proposed LCI Gathering Process

As the thesis theme suggests, product life cycle information is seen as an enabler for sustainable practices and should be gathered. The idea is to actively implement LCI collection in a structured way across all possible company units to establish an internal database. The value of information should be communicated and understood by employees to help the motivation of gathering information. The proposed LCI Gathering Process can be used as an aid to start LCI gathering in the scenario it was developed for.

It is recommended to test and establish the LCI Gathering Process across the organization with supporting systems and technologies. The process and the ways of working surrounding it should be common across all business units. This recommendation goes against the decentralized nature of the

company, but it is argued that a process could help the adoption of this new practice of gathering information. These aspects are further discussed in chapter 7.

Make the tracking label mandatory on all hose assemblies sold
To support life cycle information gathering, the existing tracking label should be put on all hose assemblies sold, even if not requested by the customer. To help customers understand this action, the case company should communicate and explain that they are taking increased responsibility for their hose assemblies and working towards enabling circular practices. Furthermore, because of the physical limitations, other label options should be explored to ensure all hoses are labelled and tracked successfully across their product life cycle. In the future, an improved label or other technologies should be included on all hoses sold. Expanding information that can be tracked in the existing tracking label system should be inspected as well, together with other systems and platforms. This is further discussed in chapter 8.

Explore systems and technologies that could support life cycle information gathering
As it was outside the thesis scope to inspect new systems or how systems can be connected, the case company is recommended to explore this topic further. This is crucial to enable information flow and unlocks data utilization. Ideally, systems should be reduced to one including all information collected by the case company and supported by tracking technologies such as an enhanced label. The feasibility of this should be further inspected.

Engage in possible circular actions that are enabled from already existing data or after new data has been collected.
Foremostly, the case company should analyse already existing data and the value of it. Over time, circular actions that are enabled with additional data should be implemented. The activities requiring minimal effort should be prioritized.

6.4 Possible Obstacles when Implementing LCI Gathering

The authors identified possible obstacles to the implementation of the LCI gathering to prepare the case company. With identified application challenges, the case company can more easily pinpoint hurdles and work proactively. A challenge identified to implement ways of working around LCI gathering is the decentralized nature of the case company. While decentralization has its advantages, the authors believe a more agile approach is suitable when working towards sustainability. Some common practices need to be established, benefiting the company time wise and economically. Accordingly, organizational culture plays a role. Motivation and attitudes towards LCI gathering may vary, making overall implementation difficult. Here it can be beneficial to showcase the benefits that come with LCI for the employees. Working together towards a common goal can feel rewarding for employees as well and encourage greater team spirit across the organization. Fostering teamwork can also improve the willingness to help one another and share knowledge. On the same topic, uneven knowledge about ways to tackle sustainability can also be a hurdle to such implementation. Therefore, it is important that there is a common understanding of why this strategy is beneficial. Some employees may also be skeptical and intimidated by change and working with a new and unclear topic such as circularity. Finally, some other big challenges are the capabilities and limitations of technologies to gather LCI. The question of system linkage and capacity is crucial. Working with user friendliness and user experience should also not be overlooked. As too complex tools may become an obstacle for LCI collection.

7 DISCUSSION

In this section, the result of the project and the methods used are discussed.

7.1 Result Discussion

The research questions have guided the work done in this project. The result discussion is done according to the research questions and compared to literature where relevant.

7.1.1 How can information gathering on products' life cycles enable circular practices for an international hose assembling company?

As stated by Jäger-Roschko & Petersen (2022), feedback in terms of life cycle information can be utilized for improving design processes to enable product disassembly and material recyclability. From the thesis work data, the authors argue that this is one of the main advantages of the LCI gathering process for enabling sustainable practices at the case company. As mentioned in interviews, the media that has been present in broken returned hoses is currently unknown, which complicates recycling. Recycling, being one of the R-strategies mentioned by Samani (2025), is a solid starting point for the case company in their transition towards circularity. Waste will always be a present issue for the case company. Hoses are a wear and tear product, no matter how well they are designed and manufactured, meaning that the case company will always have to take care of waste.

Additionally, Jäger-Roschko & Petersen (2022) argues that information enables predictable maintenance. Through the interview data and analysis, the authors see increased predictive maintenance as an opportunity to reduce the amount of waste from the case company. The LCI gathering process will provide data on maintenance occasions, which can in turn be compared to the type of hose and installation environment. By looking for correlations between these factors, the vision is to be able to estimate the lifetime of the case company products and establish predicative maintenance plans. Apart from reducing waste, it presents a business opportunity for the case company to extend the number of services provided to customers.

As stated by Saavedra-Rubio et al. (2022), data is crucial to perform an LCA study, which is one way to measure the environmental impact and performance of products. The authors believe that for a company to move and assess its journey towards circularity, data is an enabler. Not only does it unlock opportunities for circular actions discussed above, it also provides a backbone for the challenge of measuring CE mentioned by Moraga et al. (2019).

Often, as suggested in literature by Kirchherr et al. (2017), all pillars are often not considered when speaking of sustainability. Especially the social aspect can be neglected. In this thesis it has been identified that the economic pillar weighs the most for circular decisions followed by the environmental aspect. Social aspects were not explicitly brought up, except for working conditions at suppliers. This is interesting as it showcases that the social aspect of circularity seems to be more difficult to understand and define than the economic and social aspect. As the thesis focuses on environmental sustainability, it would be interesting to further explore social aspects. Note that even though the environmental factor was in focus, the economic aspect was still very present and affected decisions made in this thesis.

On the topic of sustainability, the definitions of sustainability and circularity were also found to be very diverse among interviewees. This discovery underlines the theoretical statement by Kirchherr et al. (2017) that circularity definitions vary greatly. Fitting with the literature findings by, among others,

Frishammar & Parida (2019) and Kirchherr et al. (2018), respondents also felt an increasing movement towards sustainability while still acknowledging it to be in its infancy and challenging.

7.1.2 What information gathering solutions are valuable to enable circular practices for an international hose assembling company?

The result of the project reports the LCI gathering process developed for the case company, along with the predicted values it can bring if implemented. Although, there are a couple of discussion points on its implementation possibilities.

The LCI gathering process developed is limited to a certain product journey scenario, *Private customers or small B2B purchases*, explained in section 5.2.4. The authors have discussed if and how the LCI gathering process could be implemented into the remining two scenarios, *Large customers* and *Large installations*. While it is believed that the idea of LCI gathering is something that is relevant and possible to perform in the remaining scenarios, the developed process is not directly applicable. The developed result is specifically created to fit the life cycle of products in scenario 1 as well as the working processes there and is therefore not directly translatable to other product journey scenarios. Although, a similar study as this thesis project could be performed to create a LCI gathering processes for the remining scenarios. The concept of exploring and answering the questions *what*, *how*, *who* and *when is* believed to be a good guideline to develop a new process, tailored to a different scenario or even industry. However, the authors think it would be necessary to inspect external stakeholder attitudes and collaborate more extensively with them to understand the information flow in other scenarios.

Building on this, the authors argue that the developed LCI gathering process could be partly applied to other companies or industries with a similar product offering. The type of information that is gathered, in the *what* category, is not company specific and can be used as inspiration for other companies offering hose assemblies and wish to preform LCI gathering. The other categories, *how*, *who* and *when* are tailored to the case company operations, and would therefore need to be altered if applied elsewhere. Once again, the authors believe that the concept of the developed LCI gathering process would be applicable to other companies or industries within the same field.

However, the authors argue that certain values and opportunities with life cycle information gathering identified in this project are valid both for the remining product journey scenarios as well as other similar companies. The authors have throughout the project identified that gathering life cycle information facilitates development of other circular strategies, for example waste management or designing for circularity. The more information a company has available on their products throughout their life cycle, the more accurate and reliable decisions and investigations can be made of other circular strategies.

Another topic that has been discussed in this thesis is whether to prioritize development and implementation of a process or a system first. The next step would guide the system or process. As discussed by a respondent, either option comes with its benefits and disadvantages. Through this thesis project, the authors landed in prioritizing an optimal way of working and a process to gather information to make it as user-friendly as possible, rather than inspecting a system. The authors believe that establishing a system first would cause confusion with the ways of working with the system, causing employees and users to adapt and work within the limitations of the system, potentially neglecting their needs and obstruct implementation.

7.1.3 What opportunities and risks would come with such a solution?

One risk with the LCI gathering process is the reliability of the data collected, something also discussed in literature by Sala et al. (2021) when it comes to the reliability of LCA results. The information gathering is done verbally through conversations between case company employees and service technicians. This way of gathering information works out of the assumption that the customer possesses the knowledge which is asked for and that they will provide an honest answer. The accuracy of the information collected can therefore be questioned. While some of the information gathered can be checked visually, for example which kind of media has been utilized, other information types are something only the customer knows. To avoid gathering wrongful information, the authors have recommended that it should be an option to mark questions as "unknown" in the digital platform where LCI is gathered, if the customer is unable to answer a question or is unsure. Furthermore, the authors have discarded questions which they predicted will be too difficult for customers to answer without significant effort. An example of such a question is the number of operating hours of a particular hose. For such a question, the authors suspect that the customer might give an estimated answer which is not backed up by actual data. Therefore, this question was discarded at this point in time. For these types of questions, the case company can consider utilizing sensors that are able to track such information which can be difficult for customers to answer.

While data reliability and quality when collecting data is one point of discussion, using already existing data at the case company was another topic of debate in this project. As mentioned by Kuittinen & Häkkinen (2020), already existing data should be taken advantage of. Existing data may be overseen or not refined enough to be able to use. There is also the challenge of knowing how to use data, which data and for what (Gómez-Garza et al., 2024). This was something the authors did not investigate deeper but believed to be an important aspect to create a complete flow of data. This is also crucial to avoid collecting already existing data, wasting resources such as money and time. However, the authors saw a greater need in filling the gaps, than investigating the limited amount of existing data, to enable circularity. Ideally, a database should include already existing data and currently missing information to get the best picture of products' current life cycle.

Another big challenge identified in literature (Harkonen et al., 2024; Saavedra-Rubio et al., 2022) and shown to be present in practice through the interviews at the case company, is the alignment of systems and gathering of information from multiple sources. As mentioned in section 5.2.4 and 5.3.3, a reason for choosing the scenario and focus phases and not the other scenarios was the dependence on external stakeholder information and collaboration. The authors identified challenges in gathering information from the case company's suppliers. This is also a challenge mentioned by Harkonen et al. (2024) in literature. Information flow and transparency was identified as a challenge for the case company through interviews, affecting information gathering tremendously. The existence of multiple systems containing information at the case company, along with multiple external systems used upon request from clients, also underscored the technical complexity of creating an information flow. The possibility of connecting these systems remains uncertain and is discussed in future work.

Additionally, the authors discuss the heavily decentralized nature of the company to be a potential obstacle for implementation of the LCI gathering process. In literature, Fagerlind et al. (2019) argue that both a centralized and decentralized structure have its advantages for circularity, but the most common strategy for decision-making at companies is a hybrid approach of the two. Adopting a hybrid approach allows the case company to still adapt their operations to local conditions, while decisions of sustainability strategies, which should be common across all units, will be centralized. However, as Teece (2007) argues, incumbent firms can face difficulties in altering established ways of working and integrating new practices. In order for the case company to integrate new circular practices and ways of working, such as the LCI gathering process, the authors recommend the case

company to consider a shift towards a hybrid approach of a decentralized and centralized organizational structure.

Following a change in organizational structure, the case company must investigate how to integrate new sustainability initiatives, such as the LCI gathering process, into their corporate strategy. As suggested by Baumgartner (2014), ensuring that new initiatives follow the established company sustainability vision and are translated into actionable strategies before finally being implemented on the operational level, is of importance. The authors argue that examining these management levels at the case company is important in order to reap the benefits from integrating the LCI gathering process, or other sustainability initiatives.

7.2 Method Discussion

This master thesis project has followed the triple diamond process. This process was fitting for this degree project as it allowed for deep exploration and research of both the case company and the field of circular transition before defining the most valuable area to explore further. The in-depth exploration led the authors to identify the project scope of LCI gathering, which was not apparent from the start of the project. The double diamond process was also considered; however, the selected project scope would likely have been different if applied. As the initial project brief was broadly stated, providing little indication as to where the focus of the thesis project shall lie, the exploration done in the dialogue and diagnose phases of first diamond were necessary to identify the most precise problem definition and fulfil the purpose of the thesis. Without the extensive pre-study, the authors might not have been able to identify the selected project scope. Accordingly, with the double diamond process, the problem definition would have been decided earlier in the project. While that would have left more time for developing and testing of the result, there would have been less time spent on refining the LCI process in accordance with insights gathered in the first two phases of the triple diamond. Tests for implementation of the process have now been left to future work.

The authors are therefore satisfied with the choice of the triple diamond process for the thesis project. Reflecting on how the project time was distributed for each phase, the authors in retrospect would have limited the time spent in the first diamond more. While the exploration provided valuable material, it was easy to overdo it and attempt to explore every detail of each scope that came up. To avoid this, the authors could have set up more frequent meetings to discuss and discard possible scopes to slim down the research. Setting up clear decision criteria would have facilitated such decisions.

Several methods were utilized in this project, some examples are interviews, workshops and study visits. The interviews were the main form of data collection in this thesis project. While the authors are satisfied with the number of interviews and the fact that insights were gathered from a broad range of departments, further interviews would have provided more data applicable for analysis in this project. The same reasoning goes for the study visits. As they were both done at the same case company store, insights might have been different and more nuanced if multiple stores had been visited. For analysis of the interviews, the authors utilized a codebook, explained in section 4. During analysis, it was discovered that much of the interview material was discarded as it did not fall under the category of any code in the codebook. This material is still useful for insights into the operations and sustainability work, although it did not directly connect to the research topic of LCI gathering. The workshop, which was performed together with the case company (presented in section 5.4.1) was a crucial data gathering tool. A reflection from the authors regarding such workshops is that it is challenging to lead the workshop, assist by answering questions and documenting results at the same time. Therefore, the analysis of the workshop results is solely based on the notes taken by the

participants. If documentation had been done by the authors, more nuanced and developed notes would have been taken.

7.2.1 Reliability and Validity

The main form of data collection in this thesis was semi-structured interviews. A vast majority of the interviews were held with people employed by the case company. Therefore, the authors note that there is a risk that the respondents are somewhat biased and that every opinion is subjective. The authors have attempted to avoid this issue by discussing findings with industrial supervisors at the case company continuously to confirm their accuracy and receive multiple perspectives on the findings. Additionally, continuous conversations with supervisors at Knightec Group and KTH have given the authors a variety of perspectives on the same issues, limiting the issue of bias.

8 CONCLUSION AND FUTURE WORK

This section concludes the project's results by addressing its purpose and research questions. Furthermore, future work for continued development of the result is presented.

8.1 Conclusion

In this master thesis, strategies for an international hose assembling company to transition towards circularity were explored. Following this initial brief, information was identified as a crucial enabler for circular practices, and the topic of life cycle information gathering was selected. The connection between life cycle information and circular practices was explored both in literature and at the case company through interviews and study visits. Simultaneously, benefits and challenges with information collection were identified.

The findings of this thesis and literature in the field showcase that life cycle information gathering is a challenging task. To tackle life cycle information gathering challenges, a solution was developed in the form of an LCI Gathering Process as well as connected recommendations for a sustainable transition of the case company. The idea was to assist the company to collect data in a more structured way and provide guidelines to aid the company transition towards circularity. The LCI gathering process is based on a specific product flow scenario and focuses on specific life cycle phases: *Case Company Headquarters and Warehouse, Case Company Store, Usage & Service Contact and Return to Store*. It was identified that the case company had most control to implement a process in these phases but also had the most evident responsibility to track data there. Factors affecting life cycle information gathering were also listed to create the most feasible concept possible.

The process is developed to fit current operations at the case company to make implementation easier and faster. The concept of gathering product life cycle data and seeing its value needs to be established at the company in order to transition towards circularity. The foremost obstacles to implement the process are the system and technology limitations as well as shifting the decentralized company culture. On the other hand, the process can bring environmental and economic benefits for the company by enabling circular practices such as improved waste management and increased sensitization possibilities.

Furthermore, this thesis contributes to the investigation of how life cycle information gathering can contribute to companies and their sustainability work. Additionally, it is concluded that the idea of the LCI gathering process could be implemented in other companies operating in similar industries as the case company. Other companies can take inspiration from the result, although alterations would need to be made in accordance with operations and life cycle phases of the company aiming to adopt the process. Thereby, this master thesis contributes to literature of life cycle information and its collection process, which currently is limited.

8.2 Future work

Some aspects of the result of this thesis project have been left as future work, these are presented here. Additionally, reflections on the future work of the field of LCI gathering are also presented.

8.2.1 Expansion of literature in the field

When performing the literature study for this thesis project, the authors were only able to find very limited academic literature discussing LCI gathering. The articles that are available mainly discuss information types, sharing, traceability and gathering technologies. However, the actual gathering process of the LCI seems to be a less researched topic. The literature that has been found on this topic is mainly non-academic, for example blogs or company websites discussing how they have gathered LCI. Furthermore, there is a limited amount of literature discussing the connection between circularity and LCI. The authors believe an academic literature expansion, preferably of case studies and reviews of LCI gathering, would greatly benefit the field of LCI and how it can be utilized for circular transitions and enable sustainability at companies.

Furthermore, literature of the hose assembling industry, in which the case company operates, is limited as well. While this is to be expected, as the hose assembling industry is smaller relative to for example the manufacturing industry and mainly operates on tacit knowledge, the authors noticed that literature on sustainability in assembling firms in general is limited as well. The authors recommend that expansion of literature in both topics would assist assembling companies transitioning towards circularity.

Additionally, establishing keywords for LCI research was proven difficult. Utilizing keywords such as “information” or “gathering” in academic databases provides a great number of results unrelated to the intended research field as they are words commonly used in many other contexts. Note that the authors have not conducted a systematic literature study, and therefore literature might have been overseen.

8.2.2 The LCI Gathering Process

Suggestions for technologies and digital systems for gathering LCI in practice have not been included in the results of this thesis project and are left as future work for the case company.

Regarding technologies, the case company has an established tracking label and is currently investigating other LCI gathering technologies. In the first rounds of testing for the LCI gathering process, the usage of the established tracking label is perceived to be a sufficient technology to test the process. However, it does have limitations in information types gathered currently as well as physical limitations causing it to not be suitable for all installation environments, for example in very high temperatures. Therefore, the authors recommend the case company to continue investigation of other types of LCI gathering technologies as well as continuing development of the established tracking label to increase its physical performance and track more types of information.

Regarding digital platforms for documenting and storing of the LCI gathered, future work is needed. While the authors did some quantitative analysis on which digital platform would be suitable for the LCI gathering process, it was determined that it fell outside of this project scope to investigate it in detail. The authors have instead left recommendations on what is important to keep in mind when selecting a digital platform for LCI gathering, for example that the number of digital platforms used

should be kept to a minimum to facilitate the implementation of the process. If multiple digital platforms are used, they should be able to communicate seamlessly with each other. However, the final decision of which digital platform to choose is left as future work for the case company.

The DPP is an upcoming regulation by the European Institutions which will likely affect both LCI gathering technologies and digital platforms. Possibly, which technologies and digital platforms that shall be utilized for this purpose will be decided by the European Institutions. It is therefore of importance for the case company to monitor the development of the DPP regulation to ensure their selection of technologies and digital platforms are aligned with the requirements of DPP.

The LCI gathering process has not been tested in this thesis project and is left as future work for the case company. The authors recommend the case company to begin user tests by performing pilot projects in each life cycle phase. Thereafter, evaluate the results and alternate the process accordingly. The recommended process works as a base and it is possible to build upon if new needs are discovered. The authors have left suggestions of what to consider and look for when testing and evaluating the process.

Guidelines for evaluation:

- Is the LCI gathering easy for the employee to perform?
- Does the LCI gathering take up too much time for the employees' regular work assignments? Does it fit into their work schedule?
- Are the questions intuitive for the customer? Do they understand what is asked?
- What is the customers' reaction to the LCI gathering process? Are they skeptical or positive?
- Is there any LCI information that is missing? Can further information be gathered that would benefit other circular actions?

The testing and implementation should start in Sweden with pilot projects testing the process. Each stage can be tested separately and evaluated. Changes to the process can be considered to achieve the most seamless result. The vision with the LCI gathering process is that it should be able to be expanded and continuously developed.

Further future possibilities with the LCI gathering process are to expand it to involve more stakeholders. The authors believe that once LCI gathering technologies and digital platforms are established and operating, there is a possibility to integrate, for example, customers in the gathering process as well. Granting customers access to log their products LCI information will relieve work currently put on store employees today. Additionally, the case company can offer this as a service, increasing their value offering.

8.2.3 Circular transition for case company

The recommendations made in section 6.3 are suggestions on what actions the case company should take to successfully transition towards circularity based on the study performed. The action of implementing these recommendations is future work for the case company. Once established, a re-assessment of the current sustainability state at the case company should be done to reevaluate what the next steps and strategies are to continue the transition towards circularity.

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Appendix A: Circularity Deck

This appendix shows part of the material used for a circularity deck workshop. The workshop consists of a card deck with cards suggesting circular strategies according to narrowing, slowing, closing, regenerating and informing practices. The ideas generated were placed out on the map seen in Figure 1. The card deck and more information can be found on: https://www.circularitydeck.com/

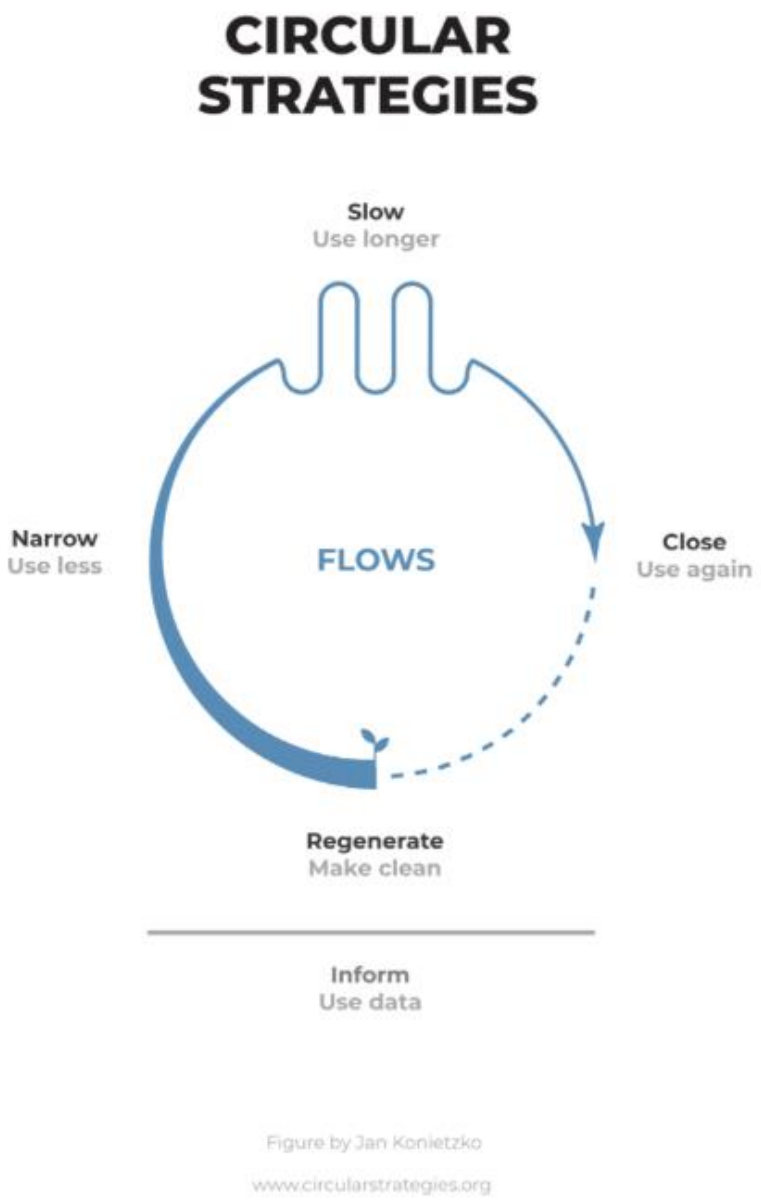


Figure 1: Work material for the circularity deck workshop.

Appendix B: Workshop instructions

The workshop material generated by the authors is presented in this appendix. The workshop was customized to the ideation phase of this thesis project, inspecting the questions what, how, who and when to be answered when collecting life cycle information. The workshop purpose and instructions are found in Figures 1-4.

WORKSHOP PURPOSE & INSTRUCTIONS

We have identified values and outlined challenges with collecting product life cycle data. **The purpose of this workshop is to brainstorm ideas around information gathering and generate possible solutions!** The ideas will help us formulate requirements for our solution.
We want your expertise, ideas and inputs for a feasible data gathering solution across all life-cycle phases.

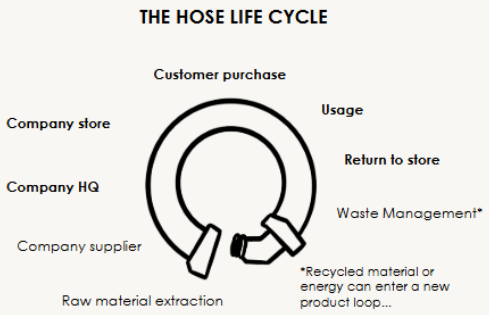
The workshop is divided in **2 parts**:

- PART 1 – The bigger picture
  - Tackling the challenge of collecting data in different phases - what's important for each phase? What do we do with already collected data and how do we collect new data?
  - Using What-Who-How?
- PART 2 – The usage phase
  - Tackling the challenge of collecting data in the usage phase (of a private customer)
  - Using How might we...?

Figure 1: Workshop purpose and overall instructions.

PART 1 – THE BIGGER PICTURE

1. Observe the life cycle of a hose for a private or a small B2B purchase.
2. Ask yourself the following questions (5 min):
   - **What** product information is interesting in each phase? Think about **what opportunities** there are with the information for your company. (~ 2 min)
   - **Who** has the product information in each life cycle phase? (~1 min)
   - **How** is it or can it be collected and tracked? (~ 2 min)
3. Discuss your ideas and thoughts for each question within your group! (5 min)
   Think about obstacles and opportunities for each topic.



IMPORTANT! KEEP ALL YOUR POST-ITS!

Figure 2: Workshop instructions for part 1.

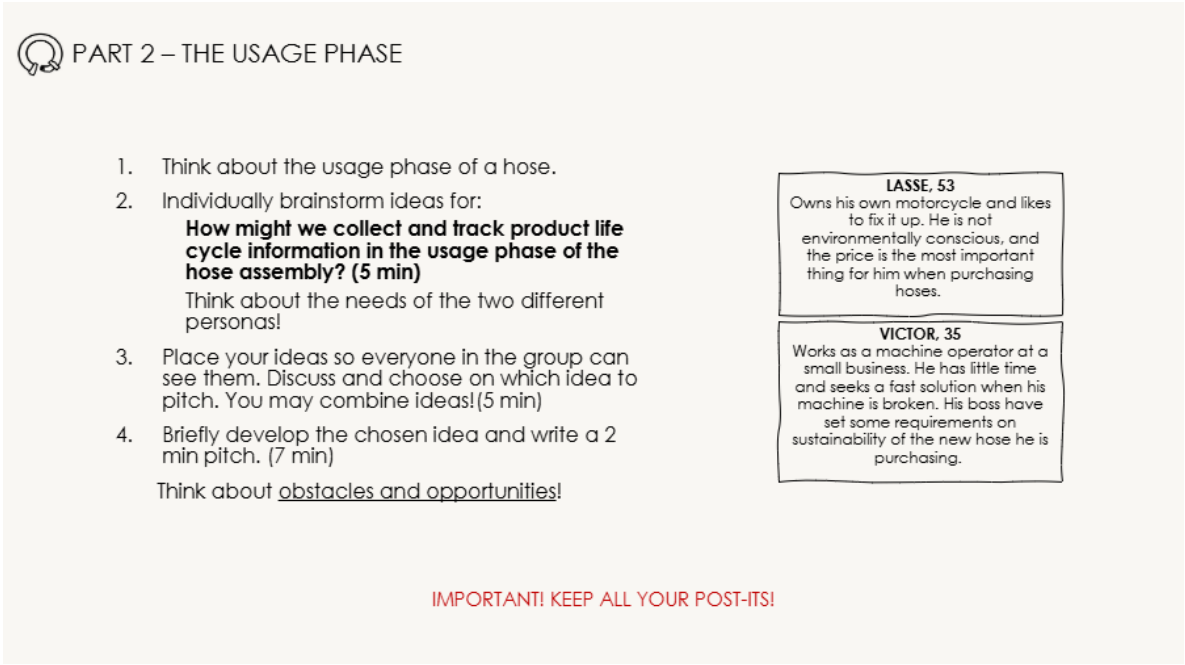


Figure 3: Workshop instructions for part 2.



Figure 4: Close up of the life cycle of a hos

Appendix C: Interview Guide

This appendix shows the common headlines of the interview guides. Note that 14 unique interview guides were generated. The interview guide presented here has been translated from Swedish.

Date & time:

Introduction of the master thesis:
We are doing this thesis in collaboration with Knightec Group, a consulting firm, from whom we got this project at the case company from. Our task is to investigate the case company's operations to identify possible opportunities for sustainability improvements in the organization. As a first step, we have investigated the case company's operations, your products and ways of working. We have identified that there are opportunities to enhance the knowledge of how your products are used by the customers through life cycle information gathering. This can, in a later stage, be utilized to adapt waste management and unlock possibilities for reuse and refurbishment of products, among other examples. The goal is to integrate sustainability as an important priority for the company and shift the mindset towards circularity instead of linearity.

In today's interview we want to ask some questions about your work as… *(interview specific)*

Formalities
- Do you want to be anonymous?
- Do we have permission to record this interview?

Introduction
- Who are you? Tell us about yourself and your role.
- What is your role at the case company?
- How long have you worked at the case company? Do you have precious experience within a similar business or industry?
- How has the company changed since you started working here? What has the journey looked like?
- How has the challenges for the company changed?
- What kind of projects have you been involved in? Anything connecting to sustainability?

Role specific questions
Here, specific questions relating to the interviewees' role were asked. Questions about product life cycle information were asked in accordance with what the respondent encounter within their role. For example, a respondent working in supply chain was asked questions of information gathering from suppliers.

Case company tracking label
- Have you ever encountered the case company tracking label? In what way?
- What is it used for? How do you use it?
- What are the benefits of it? What is not working as well?
- Is it commonly requested by customers?
- Have customers requested improvements of the tracking label?

Sustainability

- What is sustainability for you? Do you think about it in your work?
- What is sustainability for the case company? How do you define sustainability and what are your sustainability targets?
- Do you have any specific sustainability targets in your unit?
- How do you work with sustainability in your unit?
- Is sustainability requested by customers? Do they have any demands regarding sustainability?

Closure

- Based on the conversation we have had today, is there anything you would like to add?
- Have you come to think of another person you think would be beneficial for us to talk to?
- Can we get in touch with you again if we have follow up questions

Appendix D: Citations

This appendix ***showcases the original citations in Swedish, and their English translation presented in Chapter 4. The quotes are presented in the order they are mentioned in Chapter 4.***

Citation 1:
“Förut har man ju haft det i bakhuvudet när man inte kanske jobbat lika intensivt med det. Men nu är det ju egentligen hela spelplanen för oss och tänka hållbarhet hela vägen."

Translation:

“Before you’ve had sustainability in the back of your head, but maybe you haven’t worked so intensely with it. But really now it’s the entire game plan for us to think sustainability all the way”

Citation 2:
"Okej, men ska vi prata hållbarhet? Då måste vi också ha ett tänk för hur vi gör det på ett decentraliserat sätt utan att tappa liksom kärnan i vad är det vi ska uppnå."

Translation:

“Okay, if we are going to talk about sustainability, we need to find a way to do it in a decentralized manner without losing the core of what we want to achieve.”

Citation 3:
”För sälj används hållbarhetsrapporten som verktyg, det anses som ett jätteviktigt verktyg idag. Det är en dörröppnare, så det är jättebra."

Translation:

“For sales the sustainability report is used as a tool, it’s regarded to be very important tool today. It opens a lot of doors, which is very good”

Citation 4:
"Men nu har vi ändrat sätt, att vi engagerar alla andra också i hela den biten då så att de tänker på det, så att det blir ju mer en kraftansamling med teamet och inte bara att det är en person som har koll på alla arbetsmiljöer"

Translation:

“We have changed our way of working in order to engage everyone so they will think more about that part, it becomes more of a common effort from the whole team and not just one person responsible for these questions”

Citation 5:
"Vi får bara positiva återkopplingar"

Translation:

“We only receive positive feedback”

Citation 6:
"Visionen framöver är att alla produkter som vi säljer ska ha en tracking kod eller ett serienummer som vi kan följa upp"

Translation:

“The vision is that all products that we sell have a tracking label or serial number that we can follow up.”

Citation 7:
"Att om någon kommer in [i en affär] med en slang till exempel så ska de ju alltid fråga: OK, hur länge har den suttit?"

Translation:

“If someone enters [the store] with a hose it should, for example, always be asked: Okay how long has it been installed?”

Citation 8:
"I de allra bästa av världar, så skulle man ju vilja ha all information tillgängligt”

Translation:

“In the best of worlds you would want all types of information accessible”

Citation 9:
"Tracking label är ett väldigt praktiskt verktyg för oss att samla information och erbjuda kunden mer information"

Translation:

“The tracking label is a very practical tool for us to gather information and offer the customer more information”

Citation 10:
"Det är ju så att man skriver ut ett servicedokument och sen åker man till kunden och så fyller man i det på papper. Och sen kanske man då fyller i det på dator när man kommer tillbaka [till affären], eller att man då bara scannar in det där papperet."

Translation:

“It is the case that you print out a service document and then travel to the customer and fill out that paper. And then maybe you write it down in a computer when you are back [at the store] or just scan the paper”

Citation 11:
”Det hade varit värt att ha en Wikipedia för fallföretaget där man kunde skriva in någonting sådant.”

Translation:

“It would be worth it to have a case company Wikipedia where you can report such things.”

Citation 12:
”Du kan fylla i hur mycket du vill där!”

Translation:

“ You can fill out however much you like there!”

Citation 13:
"Vi ska kunna erbjuda kunder en, ja en lösning istället för att bara sälja en produkt"

Translation:

“We want to be able to provide customers a solution instead of just selling a product.”

Citation 14:
“Sen spelar det ju ingen roll hur bra system man har eller hur bra verktyg man har alltså om det blir för krångligt"

Translation:

“It doesn’t matter how good of a system you have or how good of a tool you have if it gets too complicated and troublesome.”

Citation 15:
"Folk gillar ju att prata om sina maskiner och sådant och du kan komma och lösa det och sen så går inte den slangen i sönder på fem år. Då har du en kund för livet samtidigt som du har hjälpt dem liksom"

Translation:

“People like to talk about their machines and such and if you can come and solve it and after that the hose doesn’t break for 5 years. Then you have a customer for life and at the same time you have helped them“

Citation 16:
"Vi kanske ska skapa kraven?"

Translation:

“Maybe we can create the rules?”

Citation 17:
"Hur långt ska man spåra? Vart drar man gränsen?"

Translation:

“How far should you track? Where do you draw the line?”

Citation 18:
"Det är ju önskvärt att det skulle vara så, i de allra bästa av världar, så skulle man ju vilja ha all information tillgängligt, men sen så tror jag också att blir det för mycket då. Då struntade de i det, till exempel då att om det har en checklista då trycker du bara genom samma svar på allihop om det blir för många meningar på den här checklistan.”

Translation:

“In the best of worlds you would want to have all the information available, but then I think it would be too much work. Then they might skip things on the check-list and just give the same answer for every question if there are too many questions on this check-list.”

Citation 19:
"Det gäller ju att ha med i beräkningen på alltså och hur vi lanserar det och hur vi pratar om och utbildar i det, att se till att det här är en vinning för de, för kunden och ja men för hela företaget. Det gäller ju att inspirera till att göra någonting nytt istället för att kanske tvinga på någon att göra någonting."

Translation:

“You have to keep in mind how we want to launch this and how we talk about it and train people in it, make sure that there is a win for everyone, the customer and the whole company. It’s about inspiring us to do something new instead of forcing someone to do something new.”

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